Internal Migration Restrictions, Aggregate Productivity, and Spatial Growth

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Abstract

This paper studies how institutional migration restrictions shape the long-run spatial evolution of the economy. I develop a dynamic spatial equilibrium model with endogenous migration, trade, agglomeration effects, and amenity dynamics to quantify the impact of China's *hukou* system on labor allocation, regional productivity, and aggregate economic outcomes. In the model, workers choose locations based on real income, amenities, and migration frictions; regional productivity evolves through labor-density-driven agglomeration effects; and land rents finance local amenity-enhancing expenditures through public investment.

Calibrated to prefecture-level data from China, the model reproduces key features of the Chinese spatial economy, including the concentration of economic activity in coastal metropolitan regions, persistent inland–coastal disparities, and incomplete migration despite substantial wage differentials. The quantitative results suggest that *hukou*-related migration barriers generate persistent spatial misallocation by slowing labor reallocation toward more productive regions and weakening agglomeration-driven productivity growth.

Counterfactual policy experiments indicate that relaxing migration restrictions generates substantial long-run gains in productivity, output, and welfare. Under full reform, aggregate productivity and real output increase by 2.10% and 2.89% annually over the century, respectively, while long-run consumption-equivalent welfare rises by approximately 53%. However, the gains from reform are spatially uneven. Reforms concentrated in major economic hubs yield larger efficiency gains but also intensify regional divergence and congestion, whereas broader reforms yield more balanced spatial outcomes.

More broadly, the paper highlights how migration institutions influence not only labor mobility but also the dynamic interaction between agglomeration, public-investment capacity, and regional development. While motivated by the Chinese *hukou* system, the framework provides a broader perspective on how institutional mobility restrictions shape spatial inequality, urbanization, and structural transformation in rapidly urbanizing economies.

JEL Codes: R23, R11, R13, O18, O11, R32

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1 Introduction

China's economic transformation has been accompanied by striking spatial inequality. Over the past four decades, economic activity has become increasingly concentrated in a small number of highly productive coastal and metropolitan regions, while many inland areas have experienced persistent out-migration, slower productivity growth, and weaker access to economic opportunities. At the center of this uneven spatial development lies the *hukou* system, China's household registration, which restricts migrants' access to local public services and social benefits outside their registered location (Chan & Buckingham, 2008; Song, 2014; Tombe & Zhu, 2019; Roberts, Deichmann, Fingleton, & Shi, 2012; Bosker, Deichmann, & Roberts, 2018).

The economic importance of the *hukou* system extends beyond its direct effect on migration flows. By shaping who can move, where workers settle, and how local governments finance urban development, institutional mobility restrictions influence the long-run evolution of productivity, amenities, and regional inequality. In particular, barriers to migration may prevent labor from reallocating toward more productive regions, weaken agglomeration effects, and alter the spatial distribution of land rents and public investment. As a result, migration institutions can affect not only the static allocation of labor but also the dynamic trajectory of spatial development and structural transformation.

This paper studies how institutional migration restrictions affect the long-run spatial evolution of the economy. I develop a dynamic spatial equilibrium model with endogenous migration, trade, agglomeration effects, and amenity dynamics to quantify the impact of the *hukou* system on labor allocation, regional productivity, and aggregate economic outcomes. In the model, workers choose locations based on real income, amenities, and migration frictions. Regional productivity evolves endogenously through agglomeration forces, while local amenities depend both on congestion effects and rent-financed public investment. These mechanisms generate dynamic feedback loops between migration, productivity growth, land rents, and local development.

The model incorporates several features that are central to understanding China's spatial economy. First, migration decisions are shaped by institutional barriers associated with *hukou* status, capturing migrants' unequal access to local public services and urban amenities. Second, productivity evolves endogenously through labor-density-driven agglomeration effects, allowing migration to influence future regional growth rather than merely static labor allocation (Cai, Caliendo, Parro, & Xiang, 2022; Desmet, Nagy, & Rossi-Hansberg, 2018). Third, land rents finance local amenity-enhancing expenditures, introducing a fiscal channel through which economic concentration affects infrastructure, public services, and urban livability.

Using prefecture-level data from China, the model is calibrated to match observed patterns of migration, regional productivity, trade integration, wages, and population distribution. The benchmark equilibrium reproduces several key features of China's regional economy, including the concentration of economic activity in coastal metropolitan areas, persistent inland–coastal disparities, and incomplete migration despite substantial wage differentials.

The quantitative results suggest that *hukou*-related migration restrictions generate persistent spatial misallocation by slowing labor reallocation toward more productive regions. Because productivity growth depends on labor concentration, these mobility barriers weaken agglomeration effects and reduce long-run productivity growth. At the same time, migration restrictions alter the spatial distribution of land rents and local public investment capacity, thereby affecting the evolution of amenities and urban development across regions. Counterfactual policy experiments indicate that relaxing institutional barriers to migration increases labor mobility and aggregate productivity but may also generate uneven regional outcomes, depending on the

scope and geographic design of reforms.

This paper contributes to several strands of literature. First, it contributes to the literature on spatial equilibrium and internal migration by incorporating institutional mobility restrictions into a dynamic framework with endogenous productivity and amenity evolution (Roback, 1982; Rosen, 1979; Duranton & Turner, 2011). Second, it contributes to the literature on misallocation and development by highlighting how migration barriers affect not only static labor allocation but also the dynamic evolution of regional productivity and public-investment capacity (Hsieh & Klenow, 2009; Restuccia & Rogerson, 2013). Third, the paper relates to broader debates in political economy and urban development concerning the role of state institutions in shaping spatial inequality, urbanization, and long-run economic growth (Duranton & Puga, 2012; Desmet & Rossi-Hansberg, 2014; Desmet et al., 2018; Cai et al., 2022).

More broadly, the paper emphasizes that migration institutions are not merely administrative arrangements governing population mobility. They are fundamental components of economic development strategies that shape how labor, capital, and public resources are distributed across space. In rapidly urbanizing economies, institutional barriers to mobility can therefore have persistent effects on productivity growth, regional inequality, and the trajectory of structural transformation.

The rest of the paper is organized as follows. Section 2 describes *hukou* policy and the migration constraints it creates. Section 3 presents the dynamic spatial equilibrium model. Section 4 discusses the data and calibration. Section 5 reports quantitative results and counterfactual analyses. Section 6 concludes with policy implications.

2 Institutional Context and Empirical Motivation

2.1 *Hukou* as an Institutional Friction

China's *hukou* (household registration) system, formally established in 1958, remains one of the most important institutional frictions shaping internal migration in a major economy. By tying access to local public services—including education, healthcare, subsidized housing, and pensions—to registration status, *hukou* effectively raises the cost of moving across prefectures. Even after reforms introduced temporary residency permits and, in some cities, point-based eligibility for *hukou* transfer, migrants without local registration remain excluded from many benefits. These restrictions discourage permanent settlement and create a persistent wedge between where workers are most productive and where they can fully access public services.

For the purpose of this paper, *hukou* is best understood as a recurring, location-specific migration cost. It is not a one-time expense, but a recurring institutional friction that shapes short-term mobility and long-run spatial development.

2.2 Migration Patterns under *Hukou*

Despite these restrictions, China has experienced the largest internal migration in human history. By 2020, nearly 380 million people–27% of the population—lived outside their registered *hukou* location, forming the so-called *floating population*. Flows have been highly asymmetric. Coastal metropolises such as Shanghai, Shenzhen, and Guangzhou have attracted massive inflows, while interior provinces such as Henan, Anhui, and Sichuan remain persistent outflow regions.

Migrants are disproportionately young, often with lower formal education than local urban

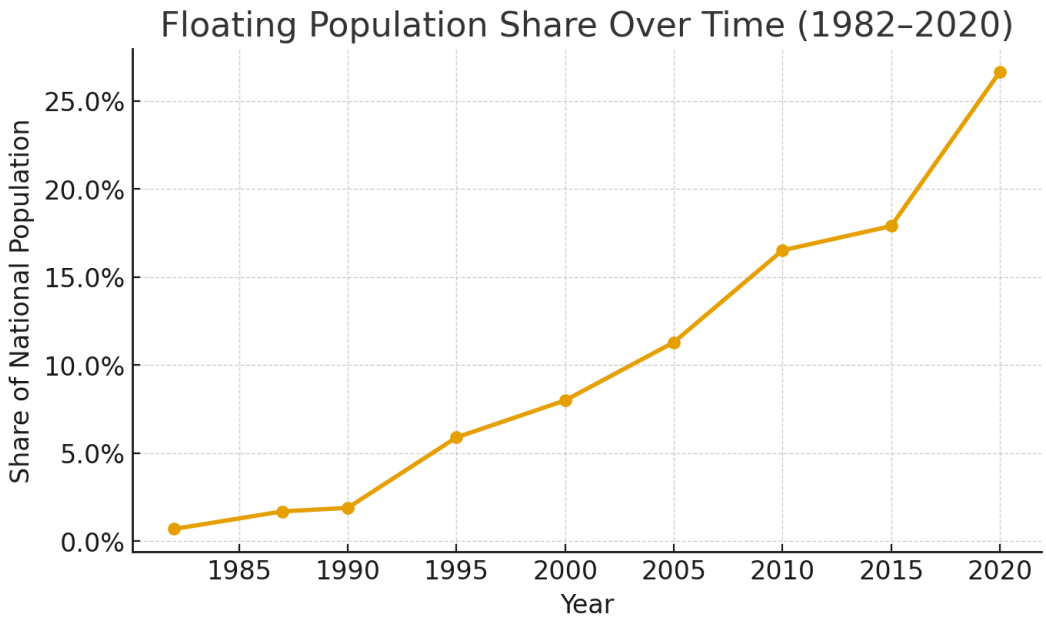


Figure 1: **Floating Population.**

residents, and concentrated in labor-intensive industries such as construction, manufacturing, and services. Their participation in the labor market has been vital to China's growth. Yet, the lack of *hukou* means they frequently occupy informal jobs and face barriers to long-term integration.

These descriptive patterns underscore *hukou*'s central role in shaping China's spatial economy. Large-scale flows occur despite high barriers, but their scale and direction are systematically conditioned by institutional rules that determine who can move there and under what conditions.

2.3 Emiprical Motivation

To illustrate the economic significance of *hukou* frictions, this subsection presents a set of stylized facts that motivate the dynamic model in Section 3.

Fact 1. Wage penalties. Migrants without local *hukou* earn substantially less than comparable *hukou* residents. Cheng, Hu, and Li (2020) estimate a 22% monthly and 32% hourly wage gap, even as rural migrants work 5.6% longer hours. A meta-analysis of 75 studies confirms that *hukou* status systematically depresses wages (Ma, Li, & Iwasaki, 2024), and survey evidence from HKUST (Wu & Zhang, 2015) finds that migrants earn about 17% less than locals after controlling for education, experience, and demographics.

Although much of this literature focuses on rural-to-urban migrants, *hukou* barriers extend to all individuals living outside their registration locality. The official category of floating population—nearly 27% of the national total in 2020—is defined as residents without local *hukou*, regardless of whether their origin is rural or urban. Since reforms after 2016 formally abolished the rural–urban *hukou* distinction but maintained locality-based restrictions, these wage penalties should be interpreted as a conservative measure of the broader disutilities faced by all cross-prefecture migrants. In other words, while rural–urban wage gaps are especially visible, the underlying mechanism is general: *hukou* exclusion generates a persistent utility wedge between locals and non-locals.

Fact 2. Limited access to services in destination cities. National survey data consistently show that migrants face barriers to public services—especially in Tier-1 cities[1] where inflows are largest. Migrant children are less likely to enroll in local public schools, while adults are less likely to receive employer-linked health insurance or qualify for subsidized housing. Thus, even as Tier-1 cities attract a growing share of the floating population, *hukou* restrictions prevent many of these households from fully integrating into local welfare systems.

In the model, these service gaps are captured as a persistent disutility wedge: migrants can work in Tier-1 destinations, but without local *hukou*, their effective utility remains below that of local residents, even when wages are equal.

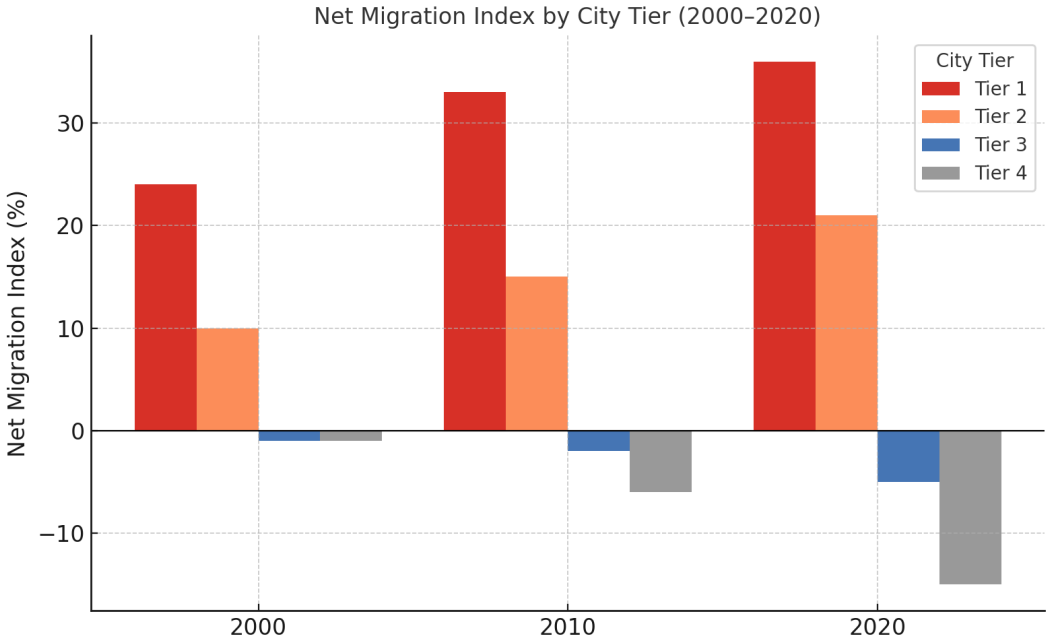


Figure 2: **Net Migration by City Tiers.**

Fact 3. Persistent regional inequality. Despite massive migration flows, spatial convergence across prefectures has been modest. Census and statistical yearbook data show diverging trends between wages and GDP per capita. Wage inequality across cities has declined steadily: the cross-city Gini fell from 0.16 in 2000 to 0.11 in 2020, and the variance of log wages also decreased. This reflects migration's equalizing effect on labor market earnings, as rural and inland workers increasingly access higher-paying jobs in urban centers.

By contrast, GDP per capita inequality has remained high. The Gini coefficient of prefecture-level GDP per capita stayed around 0.40–0.42 during 2000–2020, with only modest improvement by 2020. The variance of log GDP per capita shows a similar persistence. This discrepancy highlights a critical feature of China's spatial development: while migration narrows wage gaps, *hukou*-related barriers prevent full settlement and service access, thereby sustaining wide disparities in fiscal capacity, capital deepening, and productivity across regions.

This paradox motivates the model design. The model allows wages to adjust through labor reallocation, while persistent development inequality arises from two dynamic channels. First,

[1]I follow the GDP Tier categorization of the prefectures by South China Morning Post. All first-tier cities have a GDP over USD300 billion; the second-tier includes cities with a GDP between USD68 billion and USD299 billion; the third-tier cities have a GDP between USD18 billion and USD67 billion; the fourth-tier cities have a GDP below USD17 billion. The detailed list can be found here: https://multimedia.scmp.com/2016/cities/

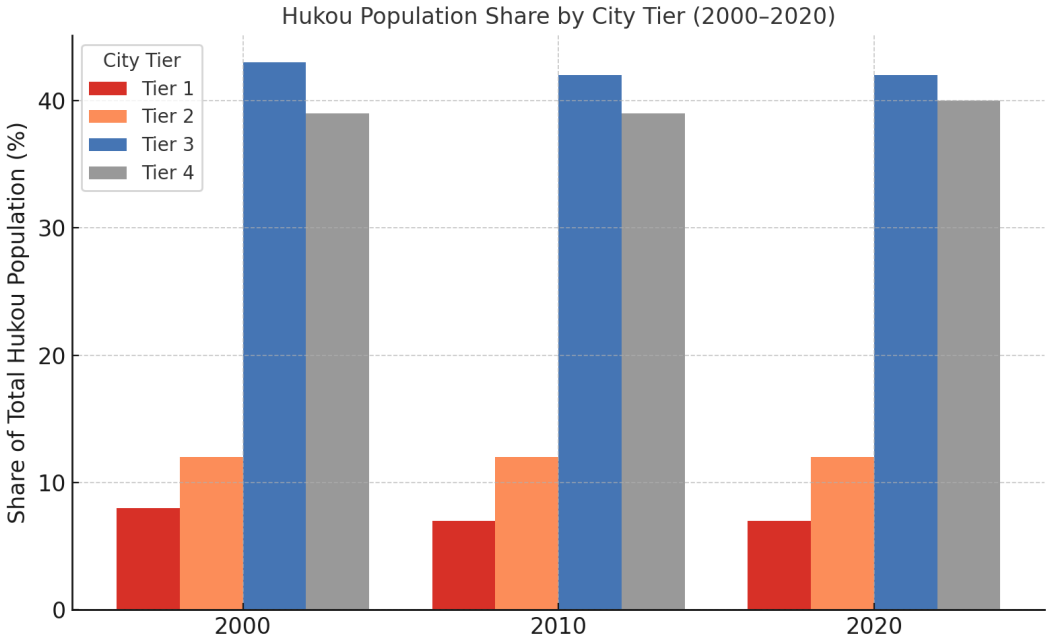


Figure 3: ***Hukou* Population by City Tiers.**

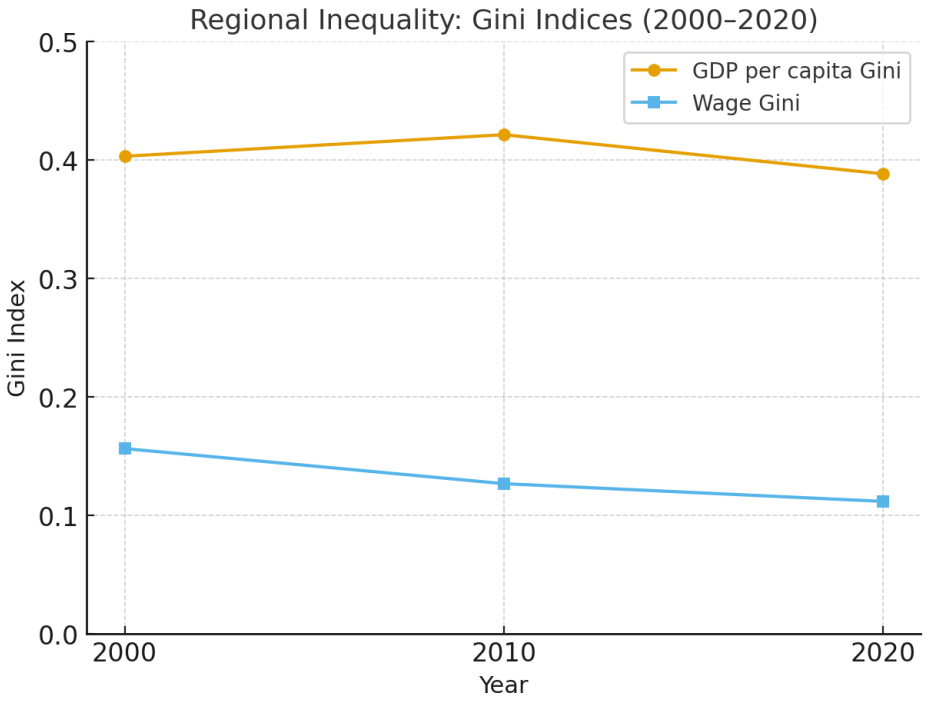


Figure 4: **Regional Inequality**

endogenous productivity dynamics strengthen regional advantages over time as agglomeration reinforces high-productivity locations. Second, endogenous amenity dynamics create competing forces: population inflows expand local fiscal capacity for infrastructure and services, yet congestion lowers livability, especially when migrants face limited access to benefits. Together, these dynamics explain why labor reallocation can narrow wage gaps while regional GDP per capita divergence remains entrenched.

Fact 4. Fragmented and Uneven *Hukou* Governance. Since the 1979 economic reforms, authority over *hukou* policy has progressively devolved from the central government to provincial, municipal, and prefectural levels. This decentralization has granted local governments substantial discretion to set admission criteria aligned with their distinct economic and demographic goals. The result is a patchwork of settlement regimes across China. Many small and mid-sized cities, often facing labor shortages or seeking to stimulate growth, have implemented relatively open settlement policies. In contrast, Tier-1 megacities have maintained highly restrictive systems. These are not uniform: for example, Shanghai employs a transparent points-based system, while Beijing uses an opaque employer-sponsored quota allocation model. Despite different mechanisms, the goal is the same: to tie settlement rights to factors such as educational attainment, skilled employment, and fiscal contributions.

Consequently, a migrant's ability to obtain local registration (and the accompanying access to public services like education, healthcare, and housing) is intensely place-specific. Significant variation exists even among cities of comparable size and economic profile; some have aggressively relaxed criteria to attract skilled labor and investment, while others retain barriers to manage population density and public resource strain. Over time, this decentralized approach has amplified regional divergence in migrant settlement opportunities.

This observed heterogeneity provides the empirical foundation for our counterfactual scenarios. The three modeled policy interventions—full national abolition (S0), targeted reforms exclusively in high-barrier cities (S1), and reforms focused on major economic hubs (S2)—are direct abstractions from the current institutional landscape. They allow us to assess whether incremental, locally-tailored reforms are sufficient to improve labor mobility and reduce welfare inequality, or if a more systemic, nationwide dismantling of the *hukou* system is necessary to unlock the full economic and social benefits of migration.

3 The Model

This section develops a dynamic spatial equilibrium model to quantify how institutional restrictions on labor mobility distort the allocation of workers across regions and, in return, affect productivity, growth, and welfare. The key friction in the discussion is the *hukou* system, which acts as a policy-induced barrier that raises the cost of migration and separates where workers are most productive from where they can access local public services.

The economy consists of N prefecture-level regions indexed by $n = 1, 2, \ldots, N$, each endowed with a fixed supply of land $H(n) > 0$. Time is discrete. The total population is $\bar{L} = \sum_h L^h$, where L^h denotes the mass of workers whose *hukou* is registered in location h. *Hukou* is fixed over time, reflecting historically low rates of status conversion before 2000.[2]

Each worker is endowed with one unit of labor and supplies inelastically in their chosen location of residence. Workers choose where to live based on real income, local amenities, and migration costs, which include both geographic frictions and *hukou*-related restrictions. These location choices determine the spatial distribution of labor, which in turn shapes local market size, productivity dynamics, and amenity formation.

Goods are traded across regions subject to symmetric iceberg trade costs $\nu(n, j) \geq 1$. In addition to domestic trade, the model incorporates international trade by allowing each prefecture to trade with an exogenous "rest of the world" (ROW) sector. This captures the fact

[2]This assumption reflects historically low rates of *hukou* change before 2000 and facilitates tractable large-scale simulations across multiple locations and periods.

that many Chinese regions are deeply integrated into global markets and rely heavily on export-oriented production and imported inputs. The ROW participates in goods markets but does not engage in migration, ensuring that the analysis remains focused on internal labor mobility and inter-prefecture trade flows while preserving tractability.

3.1 Preferences

The economy consists of a continuum of heterogeneous agents. In each period, agents choose their location of residence based on current real income, local amenities, and migration costs. Location choices are modeled as static decisions each period, while the economic environment—including wages, amenities, and productivity—evolves over time. Preferences feature idiosyncratic shocks that generate probabilistic location choices and gradual adjustment in the spatial distribution of labor.

An agent's utility is derived from the consumption of differentiated goods and the enjoyment of local amenities. Agents consume their entire income each period, so there is no intertemporal saving. For an individual i with *hukou* status h, residing in location n after moving from location j, indirect utility at time t is given by:

$$u_t^{ih}(n) = \frac{A_t(n)C_t(n)\varepsilon_t^i(n)}{\phi(n,h)m(n,j)}, \tag{1}$$

where $A_t(n)$ denotes local amenities; $C_t(n)$ is the consumption bundle or consumption index; $\phi(n,h) \geq 1$ captures *hukou*-related frictions[3]; $m(n,j) \geq 1$ represents geographic migration costs, and $\varepsilon_t^i(n)$ is an idiosyncratic location-specific preference shock which will be explained in 3.1.4.

3.1.1 Locational Amenity

Amenities are a key determinant of location choice and evolve endogenously over time. The amenity level in location n at time t is given by:

$$A_t(n) = \bar{A}(n)\left(\frac{R_{t-1}(n)H(n)}{L_{t-1}(n)}\right)^{\chi}\left(\frac{L_{t-1}(n)}{H(n)}\right)^{\lambda},$$

where $\bar{A}(n)$ captures exogenous geographic fundamentals, such as climate, natural endowments, and historical characteristics that are time invariant. The term $R_{t-1}(n)H(n)$ denotes total land rent collected in the previous period, while $L_{t-1}(n)$ is the resident population. Amenities are predetermined at the beginning of each period and depend on lagged population density and rent-financed investment, reflecting the idea that improvements in local living conditions take time to materialize.

The specification captures two distinct channels through which local conditions affect amenities. The term $\left(\frac{L_{t-1}(n)}{H(n)}\right)^{\lambda}$ reflects the impact of population density on amenities. When $\lambda < 0$, higher density generates congestion effects, such as crowding, pollution, and pressure on public services. When $\lambda > 0$, it captures agglomeration benefits, including better access to services, cultural amenities, and social interactions.

The term $\left(\frac{R_{t-1}(n)H(n)}{L_{t-1}(n)}\right)^{\chi}$ captures the role of rent-financed public investment. Higher land rents increase local fiscal resources, which can be used to improve infrastructure, environmental

[3]We should think about the costs due to *hukou* as a product of all disutility costs over time. However, this will require knowing the full history of an agent's migration, for which the data in the Census cannot provide sufficient information. Moreover, it will make the model very complicated. So, in this paper, I only consider the period disutility rather than the whole series of disutility.

quality, and public services. The parameter χ governs the effectiveness of this channel, determining how strongly increases in per capita land rents translate into improvements in amenities.

We assume that land rents are fully reinvested into local public goods, providing a link between economic activity and local living conditions. While we do not explicitly model a government, we interpret this reinvestment as occurring indirectly via the purchase of local consumption goods that improve amenities.[4]

3.1.2 *Hukou* Frictions

The parameter $\phi(n, h)$ captures the institutional barriers imposed by the *hukou* system. Each individual is endowed with a fixed place of registration h, which determines their eligibility for location-specific public services. When an individual resides in their place of registration ($n = h$), they have full access to local public goods and services, and we normalize $\phi(n, h) = 1$.

When an individual resides outside their place of registration ($n \neq h$), access to local services is restricted. In practice, migrants without local *hukou* face limited access to public education, healthcare, subsidized housing, and social insurance programs. We model these restrictions as a proportional reduction in effective utility through $\phi(n, h) > 1$, which scales down the benefits from both real income and local amenities in equation (1). Intuitively, a higher value of $\phi(n, h)$ implies that a migrant must earn higher real income or enjoy better amenities to achieve the same level of utility as a local resident.

We allow $\phi(n, h)$ to vary across destination locations n, capturing differences in the strictness of local *hukou* policies. We assume that $\phi(n, h)$ is time-invariant and individual-specific through h, reflecting the persistence of *hukou* status over the period of analysis. This formulation provides a tractable representation of institutional migration frictions while preserving heterogeneity across destinations.

3.1.3 Consumption

Agents derive utility from consuming a continuum of differentiated goods indexed by $\omega \in [0, 1]$. Preferences over goods are represented by a constant elasticity of substitution (CES) aggregator:

$$C_t(n) = \left[\int_0^1 c_t^{\omega}(n)^{\frac{\sigma-1}{\sigma}} d\omega \right]^{\frac{\sigma}{\sigma-1}},$$

where $\sigma > 1$ is the elasticity of substitution across varieties.

Agents supply one unit of labor inelastically at their chosen location, and earn wage income $w_t(n)$, which is fully spent on consumption. The corresponding price index at location n is:

$$P_t(n) = \left[\int_0^1 p_t^{\omega}(n)^{-(\sigma-1)} d\omega \right]^{-\frac{1}{\sigma-1}}.$$

Given the absence of savings, consumption equals real income, so $C_t(n) = w_t(n)/P_t(n)$ [5]. Substituting into (1) yields the indirect utility expression used in the migration decision:

$$u_t^{ih}(n) = \frac{A_t(n)}{\phi(n, h) m(n, j)} \frac{w_t(n)}{P_t(n)} \varepsilon_t^i(n), \tag{2}$$

This structure implies that location choices depend on real income, amenities, and migration frictions, with idiosyncratic shocks generating smooth labor reallocation across space.

[4] This assumption ensures that total expenditures match incomes, maintaining general equilibrium consistency without introducing an additional agent.

[5] This model does not include income taxes, though it could be extended to account for them.

3.1.4 Idiosyncratic Shocks

The idiosyncratic shocks $\varepsilon_t^i(n)$ play a central role in shaping location choices. These shocks capture unobserved, individual-specific preferences over locations, such as family ties, social networks, or personal tastes that are not explicitly modeled.

We assume that $\varepsilon_t^i(n)$ are independently and identically distributed (i.i.d.) across individuals, locations, and time, following a Fréchet distribution:

$$\Pr\left[\varepsilon_t^i(n) \leq z\right] = \exp\left(-z^{-\gamma}\right), \quad \gamma > 1. \tag{3}$$

This assumption delivers two important properties. First, it generates smooth migration flows across locations, avoiding corner solutions in which all workers concentrate in a single region. Second, it yields a tractable closed-form expression for migration probabilities, which will be derived in the next section. The parameter γ governs the dispersion of preferences: lower values imply greater heterogeneity and weaker responsiveness to differences in real income and amenities.

3.2 Labor Mobility

Labor mobility arises from individual location choices in response to differences in real income, amenities, and migration frictions. Agents are forward-looking and choose sequences of locations to maximize expected lifetime utility.

3.2.1 Location Choice

Let an individual with *hukou* status h be located in region j at time $t-1$. Given the sequence of economic fundamentals, the individual chooses a location path $n_t t \geq 0$ to maximize expected discounted utility:

$$\mathbb{E}\sum_t \beta^t u_t^{ih}(n_t),$$

where $u_t^{ih}(n)$ is given by equation (2).

The value function of worker i can be expressed as:

$$\begin{aligned} V_t^{ih}(n) &= \max_s \left[u_t^{ih}(n) + \beta\mathbb{E}\left(V_{t+1}^{ih}(s)\right)\right] \\ &= \max_s \frac{A_t(n)}{\phi(n,h)m(n,j)}\frac{w_t(n)}{P_t(n)}\varepsilon_t^i(n) + \beta\mathbb{E}[V_{t+1}^{ih}(s)], \end{aligned} \tag{4}$$

where β is the exogenous discount factor.

The probability that an agent with *hukou* status h will relocate from location n (at time t) to location s (in period $t+1$) is given by:

$$\mathbb{P}\left[V_{t+1}^{ih}(s) \geq \max_{k\neq s} V_{t+1}^{ih}(k) | V_t^{ih}(n) \geq \max_{j\neq n} V_t^{ih}(j)\right].$$

Given that the idiosyncratic amenity shocks $\varepsilon_t^i(n)$ are i.i.d. and follow a Fréchet distribution, we can substitute (2) and integrate over $\varepsilon_t^i(n)$ and obtain the probability that an individual with *hukou* h moves from location j to location n:

$$\Omega_t^h(n,j) = \frac{\left[\dfrac{A_t(n)w_t(n)}{P_t(n)} \cdot \dfrac{1}{\phi(n,h),m(n,j)}\right]^\gamma}{\sum_{k=1}^N \left[\dfrac{A_t(k)w_t(k)}{P_t(k)} \cdot \dfrac{1}{\phi(k,h),m(k,j)}\right]^\gamma}. \tag{5}$$

This formulation implies that locations with higher real income or better amenities attract more migrants, while higher geographic or institutional frictions reduce migration flows. The parameter γ governs the sensitivity of migration flows to differences in economic fundamentals.

The i.i.d. assumption of idiosyncratic preference shocks implies that the value of locating in a given region depends only on current-period utility and is independent of past location choices and future realizations of the economy. As a result, the dynamic location problem reduces to a sequence of static decisions, and agents choose their location in each period to maximize current utility.

3.2.2 Aggregation and Labor Allocation

Let $\mu_t^h(n)$ denote the share of individuals with *hukou* h residing in location n at time t. For each *hukou* type h, these shares satisfy $\sum_{n=1}^{N} \mu_t^h(n) = 1$.

Given migration probabilities, the evolution of the spatial distribution of labor is:

$$\mu_t^h(n) = \sum_{j=1}^{N} \mu_{t-1}^h(j) \Omega_t^h(n, j). \tag{6}$$

The share of individuals with *hukou* h residing in location n is given by the weighted sum of those previously residing in all origin locations j, where the weights are given by migration probabilities.

Let L^h denote the total population with *hukou* h, which is fixed over time. The number of individuals with *hukou* h residing in location n is then:

$$L_t^h(n) = L^h \mu_t^h(n). \tag{7}$$

Total labor in location n is obtained by aggregating across all *hukou* types:

$$L_t(n) = \sum_{h=1}^{N} L^h \mu_t^h(n). \tag{8}$$

3.3 Land Rents and Amenity Investment

Land is in fixed supply in each location and is used by local firms in production. Let $H(n)$ denote the fixed land endowment in location n, and let $R_t(n)$ denote the equilibrium rental price of land. Total land rent in location n is therefore $R_t(n)H(n)$.

Land is owned by immobile landlords who do not participate in the labor market. In contrast to frameworks where landlords consume goods directly, I abstract from explicit landlord consumption behavior. Instead, land rents are transferred to local governments and fully reinvested into amenity-enhancing expenditures.

This formulation captures the idea that higher local economic activity expands the fiscal or quasi-fiscal resources available for infrastructure, public services, environmental quality, and other local public goods. As a result, land rents remain part of regional expenditure and income, ensuring consistency between local income, expenditure, and goods market clearing. Moreover, in this way, migration and production shape amenities, and amenities in turn feed back into labor allocation.

Formally, this channel enters the amenity equation through rent-financed investment per resident:

$$\frac{R_{t-1}(n)H(n)}{L_{t-1}(n)}.$$

Thus, land rents do not generate a separate consumption motive for landlords; instead, they operate as a local public investment channel.

3.4 Production, Innovation, and Growth

This section outlines the structure of production at both the firm and regional levels, incorporating firm-level heterogeneity, regional agglomeration dynamics, and endogenous productivity evolution. The framework combines idiosyncratic firm-level productivity with location-specific fundamentals, allowing micro-level variation to shape aggregate outcomes in a spatial and dynamic environment.

3.4.1 Technology

A continuum of immobile firms indexed by $\omega \in [0, 1]$ in each location n produces differentiated goods. Production takes place under perfect competition and constant returns to scale, using labor and land as inputs:

$$Y_t^{\omega}(n) = Z_t^{\omega}(n) L_t^{\omega}(n)^{\iota} H_t^{\omega}(n)^{(1-\iota)}, \tag{9}$$

where $H_t^{\omega}(n)$ is land input used by firm ω; $L_t^{\omega}(n)$ is labor input; $\iota \in (0, 1)$ is the elasticity of output with respect to labor or labor share in production; $Z_t^{\omega}(n)$ denotes firm-level total factor productivity (TFP).

Productivity consists of a location-specific component and an idiosyncratic firm-level shock:

$$Z_t^{\omega}(n) = Z_t(n) \epsilon_t^{\omega}(n). \tag{10}$$

Here, $Z_t(n)$ captures the local knowledge stock and productivity fundamentals, which evolve over time through agglomeration forces, while $\epsilon_t^{\omega}(n)$ represents idiosyncratic productivity draws across firms. The dual structure allows local economic conditions to shape the distribution of firm productivity within each region.

3.4.2 Firm-Level Heterogeneity and Tractability

The idiosyncratic productivity shocks are assumed to follow a Fréchet distribution. Specifically, for each location n and period t, the shocks $\epsilon_t^{\omega}(n)$ are independently drawn from a Fréchet distribution with cumulative distribution function:

$$F(\epsilon) = \exp\left(-\epsilon^{-\delta}\right), \quad \epsilon > 0, \tag{11}$$

where ($\delta > 1$) governs the dispersion of firm productivity draws. The corresponding probability density function is:

$$f(\epsilon) = \delta \epsilon^{-(1+\delta)} \exp\left(-\epsilon^{-\delta}\right). \tag{12}$$

This implies that, in each location, a small set of highly productive draws plays a disproportionate role in determining production and trade outcomes. Importantly, the Fréchet structure delivers tractable aggregation: key equilibrium objects, including trade shares and price indices, can be expressed in closed form. This feature is essential for solving and simulating the model in a dynamic spatial setting with many locations.

3.4.3 Regional Productivity Dynamics

The location-specific productivity term $Z_t(n)$ captures the evolution of local knowledge, technology, and production efficiency. While firm-level productivity is heterogeneous within each period, regional productivity evolves endogenously over time. We model $Z_t(n)$ as an endogenous state variable that evolves over time in response to local economic conditions:

$$Z_{t+1}(n) = Z_t(n)^{\alpha} \bar{l}_t(n)^{\lambda_1} \epsilon_{t+1}(n), \tag{13}$$

where $\alpha \in (0,1)$ captures persistence in productivity, $\bar{l}_t(n)$ denotes labor density, and $\lambda_1 > 0$ governs the strength of agglomeration effects. The term $\epsilon_{t+1}(n)$ is a location-specific shock, independently distributed over time and space.

Here we captures two key forces shaping regional productivity. First, productivity exhibits path dependence, reflecting the persistence of local knowledge and technology. Second, higher labor density raises productivity through agglomeration effects, such as knowledge spillovers, learning-by-doing, and improved matching in labor markets. As workers concentrate in productive regions, these agglomeration forces generate persistent productivity advantages over time. Through this channel, migration decisions affect not only the static allocation of labor but also the dynamic evolution of regional productivity, creating a feedback loop between labor mobility and economic growth.

The stochastic component $\epsilon_{t+1}(n)$ allows for idiosyncratic local innovations or disruptions. Firm-level innovation is not modeled explicitly. Instead, firms inherit location-specific productivity, which evolves according to aggregate local conditions.

3.4.4 Firm Optimization

Firms in each location operate under perfect competition and take local factor prices as given. After drawing an idiosyncratic productivity shock, a firm producing variety ω in location n uses labor and land to produce output according to equation (12).

Given wages $w_t(n)$ and land rents $R_t(n)$, the firm chooses inputs to minimize the cost of producing one unit of output:

$$\min_{L_t^{\omega}(n), H_t^{\omega}(n)} w_t(n) L_t^{\omega}(n) + R_t(n) H_t^{\omega}(n)$$

subject to

$$Z_t^{\omega}(n) L_t^{\omega}(n)^{\iota} H_t^{\omega}(n)^{1-\iota} \geq 1.$$

The first-order conditions imply that the ratio of labor to land is given by:

$$\frac{w_t(n)}{R_t(n)} = \frac{\iota}{1-\iota} \cdot \frac{H_t^{\omega}(n)}{L_t^{\omega}(n)}. \tag{14}$$

Firms substitute between labor and land based on their relative prices, with the Cobb–Douglas structure implying constant factor shares.

$$c_t^{\omega}(n) = \frac{1}{Z_t^{\omega}(n)} \left(\frac{w_t(n)}{\iota} \right)^{\iota} \left(\frac{R_t(n)}{1-\iota} \right)^{1-\iota}. \tag{15}$$

Under perfect competition, firms make zero profits and set prices equal to marginal cost:

$$p_t^{\omega}(n,n) = c_t^{\omega}(n). \tag{16}$$

We can see that differences in productivity and factor prices translate directly into differences in production costs across locations. Regions with higher productivity or lower factor costs produce more efficiently and are more competitive in supplying goods.

3.4.5 Factor Payments and Land Rents

Given constant returns to scale and perfect competition, factor payments exhaust output. For each location n, total output is distributed between labor payments and land rents:

$$Y_t(n) = w_t(n)L_t(n) + R_t(n)H(n).$$

The Cobb-Douglas production function implies that labor and land receive fixed shares of output:

$$w_t(n)L_t(n) = \iota Y_t(n), \qquad R_t(n)H(n) = (1-\iota)Y_t(n).$$

Combining these two expressions gives the equilibrium land rent:

$$R_t(n) = \frac{1-\iota}{\iota} \cdot \frac{w_t(n)L_t(n)}{H(n)}. \tag{17}$$

Equation (17) shows that land rents rise with local wages and labor density. When migration increases the concentration of workers in a location, local production expands, and the return to fixed land rises. In the model, this matters because land rents finance local amenity investment. Thus, the rent equation creates a channel through which labor reallocation affects future amenities and, in turn, future migration decisions.

3.5 Goods Trade

Goods are traded across locations subject to iceberg trade costs. Trade links regional production and consumption decisions, allowing productivity differences across locations to shape specialization and comparative advantage.

3.5.1 Trade Costs and Delivered Prices

Shipping goods from location n to location j requires iceberg trade costs $\upsilon(n,j) \geq 1$. In order for one unit of a good to arrive at destination j, $\upsilon(n,j)$ units must be shipped from origin n.

Given unit production costs $c_t^\omega(n)$, the delivered price of variety ω produced in location n and consumed in location j is:

$$p_t^\omega(n,j) = c_t^\omega(n)\upsilon(n,j). \tag{18}$$

Substituting the unit cost function yields:

$$p_t^\omega(n,j) = \frac{\upsilon(n,j)}{Z_t^\omega(n)} \left(\frac{w_t(n)}{\iota}\right)^\iota \left(\frac{R_t(n)}{1-\iota}\right)^{1-\iota}. \tag{19}$$

Delivered prices depend on local productivity, factor prices, and bilateral trade costs. Regions with lower production costs or better market access can supply goods more competitively to other locations.

3.5.2 Regional Price Indices

Consumers purchase goods from the lowest-cost supplier across all regions. Given the Fréchet distribution of firm-level productivity, the distribution of prices can be aggregated analytically.

The CES price index in destination location j is:

$$P_t(j) = \Gamma \left[\sum_{n=1}^{N} Z_t(n)^\delta \left(\frac{w_t(n)}{\iota}\right)^{-\iota\delta} \left(\frac{R_t(n)}{1-\iota}\right)^{-(1-\iota)\delta} \upsilon(n,j)^{-\delta}\right]^{-1/\delta}, \tag{20}$$

where $\Gamma(\cdot)$ denotes the Gamma function arising from the aggregation of Fréchet-distributed productivity draws.

The price index summarizes the cost of accessing goods from all regions. Locations that are close to highly productive regions face lower prices and therefore enjoy higher real income.

3.5.3 Trade Shares

The Fréchet structure implies a closed-form expression for bilateral trade shares. The share of location j's expenditure on goods produced in location n is:

$$\pi_t(n,j) = \frac{Z_t(n)^\delta \left(\frac{w_t(n)}{\iota}\right)^{-\iota\delta} \left(\frac{R_t(n)}{1-\iota}\right)^{-(1-\iota)\delta} \upsilon(n,j)^{-\delta}}{\sum_{k=1}^{N} Z_t(k)^\delta \left(\frac{w_t(k)}{\iota}\right)^{-\iota\delta} \left(\frac{R_t(k)}{1-\iota}\right)^{-(1-\iota)\delta} \upsilon(k,j)^{-\delta}}. \tag{21}$$

Trade shares depend on relative productivity, factor costs, and trade barriers. More productive regions or regions with lower trade costs capture a larger share of expenditure in destination markets.

The parameter δ governs the dispersion of firm productivity draws and therefore the elasticity of trade flows with respect to trade costs.

3.5.4 Goods Market Clearing

Total income in location j equals labor income plus land rents:

$$X_t(j) = w_t(j)L_t(j) + R_t(j)H(j). \tag{22}$$

Goods market clearing requires that total sales by firms in location n equal total expenditure on goods produced in n:

$$Y_t(n) = \sum_{j=1}^{N} \pi_t(n,j)X_t(j). \tag{23}$$

This condition links production and demand across regions and determines the equilibrium allocation of trade flows, wages, and output throughout the economy.

3.6 Equilibrium

The equilibrium of the model consists of sequences of wages, land rents, prices, trade shares, migration probabilities, labor allocations, amenities, and regional productivity levels such that households optimize location choices, firms minimize production costs, goods and factor markets clear, and regional productivity and amenities evolve according to the model's dynamic laws of motion.

3.6.1 Static Equilibrium

Given the state variables $\{Z_t(n), A_t(n)\}_{n=1}^N$, a static spatial equilibrium at time t consists of:

$$\left\{w_t(n), R_t(n), P_t(n), L_t(n), \pi_t(n,j), \Omega_t^h(n,j)\right\}_{n,j,h}$$

such that:

1. Households choose locations optimally according to migration probabilities (Equation (5));

2. Firms minimize production costs and price at marginal cost (Equation (15));

3. Regional price indices satisfy Equation (20);

4. Bilateral trade shares satisfy Equation (21);

5. Labor allocation evolves according to Equation (6) and (8);

6. Goods markets clear (Equation (22) and (23));

7. Land rents satisfy Equation (17);

3.6.2 Dynamic Evolution

The economy evolves over time through the joint dynamics of migration, productivity, and amenities.

Regional productivity evolves according to:

$$Z_{t+1}(n) = Z_t(n)^{\alpha} \bar{l}_t(n)^{\lambda_1} \epsilon_{t+1}(n), \tag{24}$$

while amenities evolve according to:

$$A_{t+1}(n) = \bar{A}(n) \left(\frac{R_t(n) H(n)}{L_t(n)} \right)^{\chi} \bar{l}_t(n)^{\lambda}. \tag{25}$$

These equations generate a dynamic feedback mechanism linking migration, production, land rents, and local amenities. Migration reallocates labor across space, affecting productivity growth and land rents. In turn, changes in productivity and amenities alter future real income and migration incentives, shaping the long-run spatial distribution of economic activity.

3.6.3 Existence and Uniqueness of Equilibrium

To ensure an equilibrium exists and is unique, we follow the analytical approaches of Allen and Arkolakis (2014) and Desmet and Rossi-Hansberg (2014). In particular, the uniqueness of the equilibrium of the model depends on the relative strength of agglomeration forces (which attract labor to dense regions) and dispersion forces (which push labor away due to congestion and diminishing returns).

The model yields a unique static equilibrium at each time period if the following condition holds: The sum of agglomeration elasticities from productivity spillovers, amenity investment, and population density effects must not exceed the sum of dispersion forces from production congestion, amenity crowding, migration frictions, and trade substitutability.

Mathematically, this is expressed as:

$$\frac{\lambda_1}{\delta} + \chi < |\lambda| + (1 - \iota) + \frac{1}{\gamma},$$

where

- $\frac{\lambda_1}{\delta}$: strength of productivity agglomeration from labor density;
- χ: elasticity of amenities to rent-driven investment;
- λ: elasticity fo amenities to labor density;
- $(1 - \iota)$: dispersion force from decreasing returns to labor in production;

- $1/\gamma$: migration dispersion due to idiosyncratic preference shocks.

If this inequality is satisfied every period, then the equilibrium allocation of labor, production, and prices is unique. If not, the model may exhibit multiple equilibria, where small differences in initial conditions lead to persistent divergence across regions.

4 Quantitative Analysis

This section presents a quantitative analysis of the dynamic spatial model using prefecture-level data from China. The main objective is to calibrate and estimate the structural parameters of the model, discipline key mechanisms using micro and macro data, and use the resulting framework to simulate the spatial distribution of labor, wages, productivity, and migration dynamics in the presence of *hukou*-induced mobility frictions. The model features Fréchet distribution firm productivity, land rents funding amenity investments, and free entry and exit in the production sector, generating endogenous agglomeration and dispersion forces.

The analysis proceeds in four steps. First, we describe the data sources used to construct key model variables. Second, we outline the empirical strategies used to estimate or calibrate structural parameters. Third, we derive initial conditions such as productivity and amenities for the starting year. Finally, we compute compound mobility frictions and decompose them into geographic and institutional components.

4.1 Data

The quantitative analysis uses prefecture-level data from China to construct a dynamic spatial economy with endogenous migration, trade, productivity, and amenity dynamics. The spatial units consist of 313 prefecture-level cities, harmonized over time to account for administrative boundary changes and missing observations. The benchmark year for calibration is 2000, which predates major nationwide *hukou* reforms and provides a natural baseline for evaluating long-run spatial dynamics.

Population, migration, and labor market information are primarily drawn from the 2000, 2010, and 2020 China Population Censuses. These datasets provide information on resident population, *hukou* registration, migration status, employment, and wages. Because the census records both place of residence and place of *hukou* registration, it allows the construction of inter-prefecture migration flows under institutional mobility restrictions.

To supplement the census, I use the China Migrants Dynamic Survey (CMDS), which provides detailed information on migrant characteristics, migration histories, wages, *hukou* status, and destination choices. These data are used primarily to estimate migration elasticities and document the magnitude of *hukou*-related mobility frictions.

Regional economic variables, including GDP, employment, wages, land use, and sectoral output, are obtained from the China City Statistical Yearbooks. These data are used to construct measures of regional productivity, labor income, population density, and land rents. Data on local amenities, including environmental quality, healthcare capacity, educational resources, and public infrastructure, are compiled from the China City Statistical Yearbooks and official statistical bulletins.

Exogenous geographic fundamentals are constructed using climatic and topographic variables obtained from the China Meteorological Data Service Center and the National Geographic Information Center. These variables include temperature, precipitation, humidity, elevation,

and terrain ruggedness, and are used to identify the time-invariant component of local amenities.

Bilateral trade costs are constructed using GIS-based travel times between prefectures. Following the spatial trade literature, I compute the shortest travel time between each prefecture pair using the ArcGIS Origin–Destination Cost Matrix tool applied to China's transportation network in 2000, including highways and major railways. These travel times are then transformed into iceberg trade costs used in the quantitative model.

4.2 Parameter Estimation and Calibration

The model is parameterized using a combination of externally calibrated values, direct estimation, and model inversion. Parameters are chosen to match key features of the Chinese spatial economy, including migration responses, regional productivity dynamics, trade integration, and amenity variation across prefectures. Table 1 summarizes the baseline parameterization.

4.2.1 Externally Calibrated Parameters

Several parameters are assigned standard values from the literature or calibrated directly from long-run macroeconomic conditions.

The discount factor β is set to 0.776, corresponding to an annual interest rate of approximately 2.5% over a ten-year model period. The elasticity of substitution across differentiated goods, σ, follows standard values in the quantitative trade literature. This parameter determines the responsiveness of expenditure allocation to relative prices and affects the strength of market competition across regions.

The Fréchet shape parameter δ, which governs the dispersion of firm-level productivity dispersion and the sensitivity of trade flows to trade costs, is set to match standard values used in the Eaton–Kortum trade literature[6]. The persistence parameter in regional productivity dynamics, α, governs the degree of path dependence in local productivity evolution over time[7].

4.2.2 Estimated Structural Parameters

The remaining parameters are estimated or calibrated to match moments from Chinese prefecture-level data.

The labor share parameter ι governs the importance of labor relative to land in local production:

$$Y_t^{\omega}(n) = Z_t^{\omega}(n) L_t^{\omega}(n)^{\iota} H_t^{\omega}(n)^{1-\iota}.$$

I estimate ι using prefecture-level data on labor compensation and regional output from the China City Statistical Yearbooks.

The migration elasticity parameter γ governs the responsiveness of migration flows to differences in real income, amenities, and migration frictions across regions. Higher values of γ

[6]Caliendo, Dvorkin, and Parro (2019) find that trade elasticity typically falls within the range of 3 to 8, which is consistent with earlier estimates by Eaton and Kortum (2002) and Head and Mayer (2014). However, intra-country trade generally faces smaller barriers compared to international trade, so the trade elasticity δ in this model is calibrated at 4.55, towards the lower end of the literature's spectrum.

[7]This parameter captures the concept of "path dependence" or "productivity persistence," where regions with a history of high productivity are more likely to maintain high productivity levels in the future due to accumulated advantages like knowledge, continuous innovation, and superior infrastructure. This value is based on studies by Allen and Donaldson (2020), Moretti (2012), Comin and Hobijn (2010), Fagerberg, Srholec, and Knell (2007) and many others, making 0.98 a reasonable value.

imply stronger migration responses to spatial differences in utility. The parameter is estimated using bilateral migration flows constructed from census and CMDS data.

The agglomeration elasticity λ_1 captures the effect of labor density on regional productivity growth:

$$Z_{t+1}(n) = Z_t(n)^{\alpha} \bar{l}_t(n)^{\lambda_1} \epsilon_{t+1}(n)$$

.

This parameter governs the strength of agglomeration forces through mechanisms such as learning spillovers, knowledge diffusion, and local scale effects. I estimate λ_1 using panel regressions relating regional productivity growth to lagged labor density across prefectures.

Amenities evolve through both congestion effects and amenity-enhancing public investment financed by land rents. The amenity elasticity λ captures the effect of population density on local livability, while χ governs the elasticity of amenities with respect to rent-financed amenity investment. These parameters are estimated using a two-step procedure that decomposes observed amenities into exogenous and endogenous components before estimating the contribution of density and public-investment channels.

Trade costs are modeled as symmetric iceberg costs between prefectures:

$$v(n, j) \geq 1$$

.

Bilateral trade costs are constructed from GIS-based travel times across China's transportation network and transformed into iceberg trade barriers following standard specifications in the quantitative trade literature.

Migration costs combine geographic mobility frictions with institutional barriers generated by the hukou system. Geographic migration costs depend primarily on bilateral distance and travel time, while institutional migration frictions vary systematically across destinations according to local hukou restrictions and access to public services.

Additional details on estimation procedures, parameter identification, and robustness checks are reported in Appendix D.

Model Parameters. Below is a summary of key parameters used in the model, along with their sources and justifications:

Table 1: **Baseline Parameter Values.**

Parameter	Description	Value	Source / Notes
β	Discount factor	0.776	Implied by a 2.5% annual interest rate over a decade
σ	Elasticity of substitution across varieties	4.00	(Krugman, 1991)
δ	Trade elasticity	4.55	(Caliendo et al., 2019)
α	Persistence of regional productivity	0.98	(Allen & Donaldson, 2020)
λ	Elasticity of amenities w.r.t. population density	-0.39	GMM estimate from urban amenity regression
χ	Elasticity of amenities w.r.t. amenity investment	0.23	GMM estimate from urban amenity regression
γ	Elasticity of migration flows w.r.t. real income	2.00	Estimated from migrant flows and wage gaps
λ_1	Agglomeration elasticity in productivity dynamics	0.21	Estimated from density–productivity regressions
ι	Labor share in production	0.58	Estimated using prefectural GDP and wage data

Notes: GMM = Generalized Method of Moments. "w.r.t." denotes "with respect to." All estimates are at the prefecture level unless otherwise noted.

4.3 Initial Conditions and Model Inversion

The quantitative analysis requires recovering the set of regional fundamentals that rationalize the observed benchmark spatial equilibrium in the base year. Given observed data on wages, population allocation, migration flows, and trade costs, the model is inverted to recover regional productivity, exogenous amenities, and migration frictions consistent with equilibrium behavior. These recovered fundamentals serve as the initial conditions for the counterfactual experiments.

4.3.1 Recovering Regional Productivity and Exogenous Amenity

Given initial population $\bar{L_0}(n)$, wage w_0, and land from the data, we begin by recovering initial regional productivity $Z_0(n)$ from the model's equilibrium wage equation:

$$w_0(n) = (1-\iota)^{-1}\iota^{\frac{\chi-1}{\chi}}\left[\frac{\tilde{u}_0(n)}{\bar{A}(n)}\right]^{\frac{1}{\chi}} Z_0(n)^{-\frac{1}{\chi}}\bar{l}_0(n)^{\frac{1-\iota-\lambda}{\chi}}, \tag{26}$$

where $\tilde{u}_0(n) = A_0(n)\frac{w_0(n)}{P_0(n)}$ is the deterministic component of utility excluding idiosyncratic preference, amenity loss, and migration cost.

Rearranging the equation allows us to express initial productivity:

$$Z_0(n) = (1-\iota)^{-\chi}\iota^{\chi-1}\frac{\tilde{u}_0(n)}{\bar{A}(n)}w_0(n)^{-\chi}\bar{l}_0(n)^{1-\iota-\lambda} \tag{27}$$

To implement this step, we must obtain the ratio $\frac{\bar{A}(n)}{\tilde{u}_0(n)}$ for each location. From the price index formulation derived in the trade block of the model, we obtain the following equation:

$$\begin{aligned} &\left[\frac{\bar{A}(n)}{\tilde{u}_0(n)}\right]^{-\delta} w_0(n)^{-\delta(\chi+1)} \bar{l}_0(n)^{-\lambda\delta} \\ &= \kappa \left[\frac{\iota}{1-\iota}\right]^{2\chi\delta} \sum_j \left[\frac{\bar{A}(j)}{\tilde{u}_0(j)}\right]^{-\delta} T(j) w_0(j)^{-\delta(1+\chi)} \bar{l}_0(j)^{-\lambda\delta} \upsilon(n,j)^{-\delta}, \end{aligned} \tag{28}$$

where $\kappa = \left[\Gamma(\frac{1-\sigma}{\delta}+1)\right]^{\frac{\delta}{\sigma-1}}$ is a constant.

This expression enables us to recover the ratio $\frac{\bar{A}(n)}{\tilde{u}_0(n)}$ up to a normalization, given observed wages, population, and estimated bilateral trade costs. Then, we return to the productivity equation to recover the initial productivity $Z_0(n)$ for each location.

To pin down the exogenous amenities $\bar{A}(n)$ directly, we use the consumption-based proxy for utility $\tilde{u}_0(n)$ with data on per capita consumption expenditures in 2000. This allows us to compute $\bar{A}(n)$ from the estimated ratio.

This approach ensures consistency between the model's equilibrium conditions and the observed spatial distribution of wages, population, and trade, while avoiding strong assumptions about future growth paths. Full derivations and solution details are provided in Appendix A.

4.3.2 Mobility Costs and Amenity Lost

To implement the model, it is crucial to estimate mobility costs across locations. We use four sets of inputs: (i) population distributions for periods 0 and 1, (ii) *hukou* population by location for periods 0 and 1, (iii) observed migration flows between periods 0 and 1, and (iv) the exogenous amenity level $\bar{A}$ recovered in the previous subsection.

From equation (5), the bilateral migration probability of *hukou* type h moving into location n from origin s in period 0 is

$$\Omega_0^h(n,s) = \frac{\tilde{u}_0(n)^\gamma, \tilde{m}(n,s)^{-\gamma}}{\sum_{j=1}^N \tilde{u}_0(j)^\gamma, \tilde{m}(j,s)^{-\gamma}}, \tag{29}$$

where $\tilde{m}(n,s) = m(n,s)\phi(n,h)$ is the compound migration cost, consisting of distance-related cost $m(n,s)$ and the *hukou*-related penalty $\phi(n,h)$.

Consider the scenario where an individual remains in their registered *hukou* location ($n = h$), implying that $\tilde{m}(n,n) = 1$, since both the *hukou* parameter and movement cost equal one. The staying probability (remaining in one's registered *hukou* location) implies

$$\Omega_0^{h=n}(n,n) = \frac{\tilde{u}_0(n)^\gamma}{\sum_{j=1}^N \tilde{u}_0(j)^\gamma, \tilde{m}(j,n)^{-\gamma}}. \tag{30}$$

Taking the ratio of equations (26) and (27), and rearranging, we can derive a closed-form expression for the compound mobility cost from the data:

$$\tilde{m}(n,s) = \left[\frac{\Omega_0^{h=n}(n,n)}{\Omega_0^h(n,s)}\right]^{1/\gamma}. \tag{31}$$

Using the locational productivity derived for period 0, $Z_0(n)$ and the labor population data $\bar{l}_0(n)$ along with the migration flows of *hukou* population, we can calculate mobility cost $\tilde{m}(n,s)$ for each location from the model. Detailed computations are provided in the Appendix A.

To isolate the amenity loss due to *hukou* from the compound migration cost $\tilde{m}(n,s)$, I model the migration cost based solely on distance, denoted as $m(n,s)$. This distance-based migration cost is assumed to be symmetric between two locations, reflecting the idea of an invariant baseline migration cost. It incorporates aspects such as the loss of social networks and capital at the place of origin, as well as the psychological toll of leaving family and familiar surroundings. However, it may not fully account for all tangible costs associated with moving and settling in a new area.

To discipline the elasticity parameter θ of the distance-based migration cost, I utilize data from the CMDS (China Migration and Demographic Survey). The following gravity model is constructed to align with observed data on population, distance, and migration flows:

$$M(n,s) = G\frac{P_n^{\nu_1}P_s^{\nu_2}}{D(n,s)^{\theta}},$$

where G is a constant, P_n and P_s represent the populations of locations n and s, respectively, and ν_1 and ν_2 are the elasticities of migration flows with respect to the populations at the origin and destination. $D(n,s)$ denotes the distance between locations n and s, and θ is the elasticity of migration flows with respect to distance.

From this model, I estimate $\theta = 1.32$, which aligns with the typical range of 0.7 to 2 found in the literature. This estimated migration cost allows us to isolate the "pure" frictions associated with *hukou* from the "compound" migration cost.

4.4 Counterfactuals Analysis: Removing Institutional Barriers

Having fully parameterized and calibrated the model using historical data and empirical estimates, I proceed to analyze the long-run spatial implications of institutional barriers to labor mobility—China's *hukou* system. The objective of this counterfactual exercise is to isolate the dynamic consequences of lifting migration frictions while holding other structural features of the economy constant.

4.4.1 Simulation Approach

The counterfactual is implemented using the dynamic hat algebra method, which facilitates tractable analysis of long-run spatial transitions by computing relative changes in endogenous variables over time rather than solving for their levels directly. The model is initialized with observed data in period $t = 0$, reflecting the status quo in which the *hukou* system continues to distort individual location choices through migration frictions embedded in the bilateral cost function, $\phi(n,h) > 1$, for migrants without local registration.

I then simulate the spatial evolution of the economy under three scenarios:

1. **Baseline.** This scenario maintains the existing *hukou* system throughout the transition, with all bilateral migration frictions remaining unchanged over time.

2. **Full reform.** This scenario abolishes the *hukou* system from period $t = 0$ onward by setting $\phi(n,h) = 1$ for all agents and destinations. This implies that all workers face equal amenity costs regardless of their origin, thereby removing institutional barriers that previously restricted mobility across regions.

3. **Selective/partial reform.** This scenario abolishes the *hukou* system only in designated reform prefectures. Formally, $\phi(n,h) = 1$ if n is a reform prefecture, while $\phi(n,h) > 1$

elsewhere. This captures the reality of partial reforms in which migrants gain local access rights in some destinations but not nationwide.

All scenarios are simulated for 10 periods (interpreted as 50 to 100 years), allowing the economy to reach a new long-run spatial equilibrium. In each scenario, I solve for the full set of endogenous equilibrium objects over time, including regional utility levels $u_t(n)$, labor distributions $\bar{l}_t(n)$, and productivity levels $Z_t(n)$.

4.4.2 Interpretation and Outcomes

By comparing the paths $\hat{u}_t(n)$, $\hat{\bar{l}}_t(n)$, and $\hat{Z}_t(n)$ from the counterfactual scenario to those under the benchmark, I assess the dynamic effects of removing institutional frictions on the spatial distribution of population and productivity. The key outcomes of interest include:

1. **Labor distribution.** The reallocation of labor across regions, highlighting areas that gain or lose population once migration constraints are lifted;
2. **Productivity.** Changes in regional productivity, which evolve endogenously through agglomeration effects and innovation dynamics; and
3. **Welfare.** Welfare improvement is computed as changes in lifetime utility aggregated across individuals and weighted by the initial population distribution.

To maintain a clear empirical focus, this analysis restricts attention to the effects of institutional reforms alone. In particular, improvements in transportation infrastructure—such as the expansion of high-speed rail (HSR)—are excluded from this chapter. These infrastructure changes often co-evolve with migration and productivity, but they constitute a distinct set of policy instruments with their own modeling complexities. I therefore reserve the analysis of HSR and related transportation improvements for a separate chapter, where I will explicitly model trade costs, access, and infrastructure-induced migration incentives in interaction with institutional reforms.

Quantitative results and visualizations of the key counterfactual trajectories are presented in Section 5.

5 Quantitative Results: Mobility, Productivity, and the *Hukou* System

This section presents the results of the quantitative analysis introduced in Section 4, focusing on the role of the *hukou* system in shaping internal migration, productivity, and spatial development in China. We begin by examining the model's calibrated initial conditions, which capture the economic geography at the turn of the 21st century. We then turn to the estimated migration frictions—including institutional constraints—and assess how well the model reproduces key features of the data. Finally, we conduct a counterfactual analysis to explore the long-run effects of abolishing the *hukou* system on labor reallocation, regional productivity, and aggregate welfare.

5.1 Initial Spatial Economic Conditions

The model begins in 2000 and incorporates key features of China's economic geography: exogenous amenities and initial productivity. These factors collectively shape the initial distribution of population and economic activity and help explain observed migration patterns prior to major reforms.

5.1.1 Exogenous Amenities

Figure 5 maps the estimated exogenous amenities $\bar{A}(n)$ across prefectures, reflecting geographic and environmental endowments. Contrary to the assumption that coastal areas uniformly offer the highest amenities, the map reveals a more nuanced pattern.

Some inland regions — such as Sichuan, Yunnan, and Guangxi — exhibit relatively high amenity values, driven by climate, topography, and natural beauty. These regions attract labor less through industrial agglomeration than through environmental quality. Coastal cities in Guangdong and Zhejiang also feature high amenity levels, reflecting fertile land, trade access, and historical investment.

By contrast, the western provinces (e.g., Tibet, Qinghai, Xinjiang) and parts of Inner Mongolia show lower amenity levels, consistent with harsh terrain, limited arable land, and weaker infrastructure. The northeast industrial belt (Liaoning, Jilin, Heilongjiang) shows moderate amenities, reflecting a mix of industrial legacy and declining environmental quality.

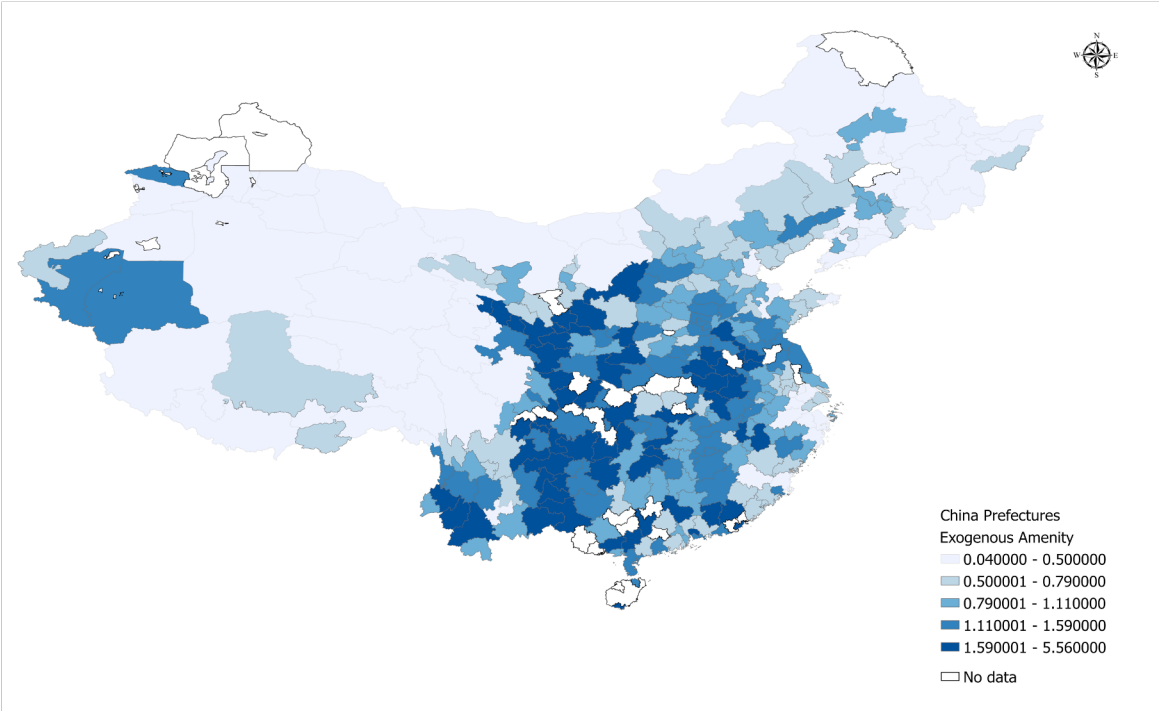


Figure 5: **Exogenous Amenities by Prefecture.**

5.1.2 Initial Productivity

Figure 6 displays initial total factor productivity (TFP) across prefectures in 2000. The highest levels are concentrated in the Yangtze River Delta, Pearl River Delta, and Bohai Economic Rim, encompassing cities such as Shanghai, Guangzhou, and Beijing. These areas benefited

from early liberalization, foreign investment, and policy support, including Special Economic Zones. Their integration into global supply chains and proximity to export markets sustained high levels of industrial output and labor demand.

Productivity levels in central and southwestern regions vary widely, reflecting mixed development paths. Provinces like Sichuan show moderate productivity, supported by local manufacturing and a growing service sector. More remote inland areas remain lagging due to weaker industrial bases and transportation constraints.

The northeast, historically industrialized under the centrally planned economy, maintains moderate to high productivity in some prefectures. However, the onset of deindustrialization and state-owned enterprise reform in the 1990s led to economic stagnation and rising unemployment in parts of the region, contributing to population outflows and declining investment.

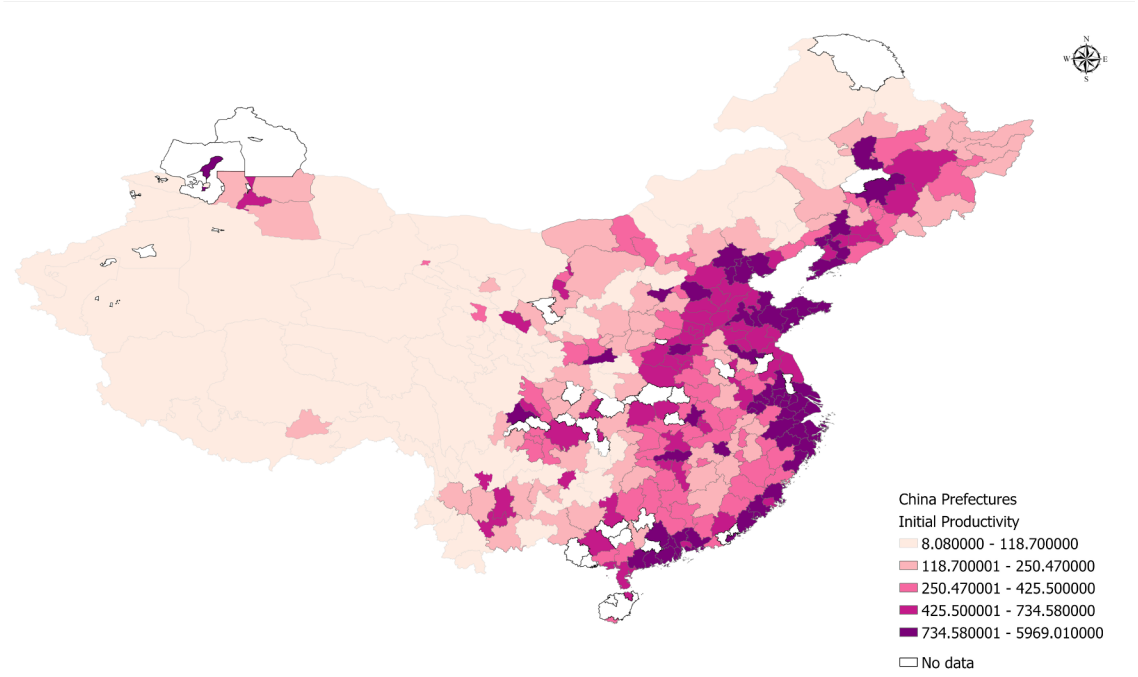


Figure 6: **Prefecture-Level Productivity in 2000.**

5.1.3 Initial Utility and Wage Distribution

To further characterize regional heterogeneity, the model calibrates regional deterministic utility and wage levels across locations in the initial period. These outcomes are shaped jointly by productivity, amenities, trade costs, and migration frictions. In equilibrium, they summarize the economic and social value of each location from a resident's perspective.

The left panel of Figure 7 displays the spatial distribution of real wages (in units of 1,000 RMB), computed by the model based on estimated productivity, exogenous amenities, and the equilibrium labor distribution. The highest wages are concentrated in coastal cities such as Shenzhen, Guangzhou, Foshan, and Zhuhai in Guangdong province; Suzhou and Wuxi in Jiangsu; and Xiamen in Fujian. Beijing also stands out, benefiting from its administrative and political centrality. Resource-intensive cities like Karamay, Dongying, Daqing, and Panzhihua show elevated wage levels, primarily due to their strong presence in extractive industries. These patterns reflect the combined influence of firm-level productivity, sectoral composition, and

enhanced market access in these locations.

In contrast, western and some central inland regions show considerably lower real wages, reflecting both lower productivity and limited market access. The relative isolation of these areas — due to geography and underdeveloped infrastructure — restricts the size of local markets and firms' ability to scale, thereby constraining wage growth.

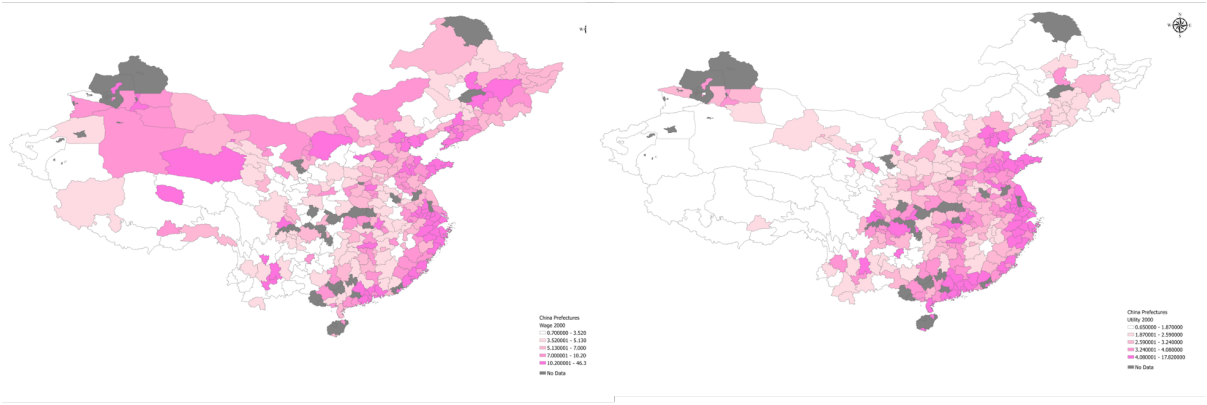


Figure 7: **Model-Implied Wage vs. Utility in 2000.**

The right panel of Figure 7 presents the initial utility, which incorporates both consumption and amenity components. The spatial variation in utility reflects not only trade-offs between economic opportunity and quality of life but also the reinforcing effect of high wages and high amenities in many leading cities. In particular, several top wage locations, such as Shenzhen, Guangzhou, and Suzhou, also rank among the highest in deterministic utility, suggesting that these places provide both strong economic returns and favorable living conditions, even after accounting for congestion and baseline frictions. This reflects a close alignment between wage and utility rankings across most regions. However, there are important exceptions. Cities like Karamay, Daqing, and Panjin exhibit high wage levels but do not rank similarly in utility. This discrepancy may be attributed to lower exogenous amenities or higher consumption goods prices due to trade frictions and geographic isolation, which reduce the effective purchasing power and desirability of these locations despite high nominal wages. Some inland cities offer relatively high utility despite modest wage levels, driven by favorable amenities, lower congestion, or lower costs of living. Meanwhile, high-income coastal cities may exhibit only moderate utility once crowding factors are accounted for.

Utility disparities across prefectures signal the presence of significant spatial frictions. While economic forces would predict greater labor reallocation toward high-productivity regions, institutional barriers — notably the *hukou* system — prevent full equalization of utility across space.

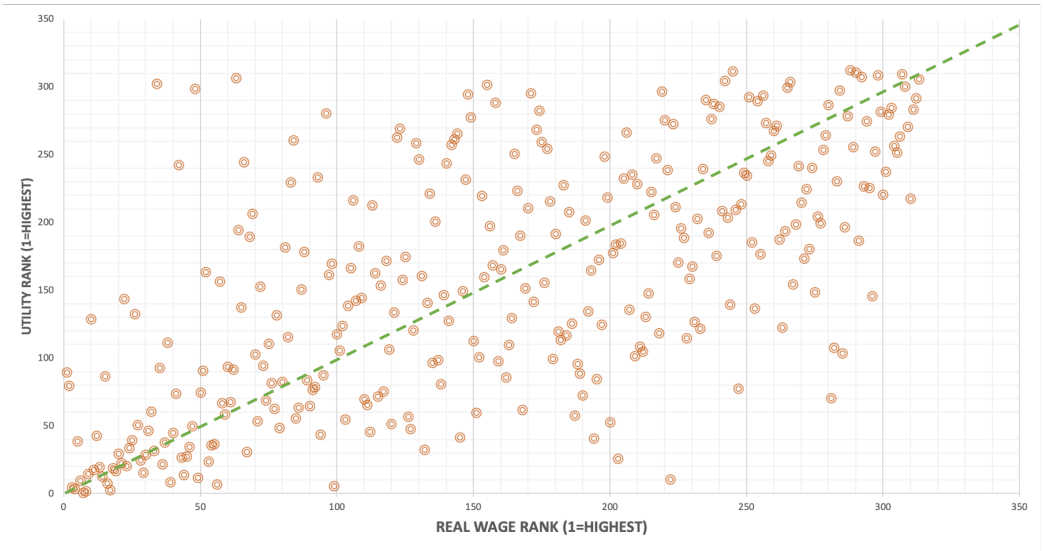


Figure 8: **Real Wage and Utility in 2000.**

Overall, these initial conditions provide a consistent backdrop for analyzing subsequent mobility decisions and policy counterfactuals. Regional disparities in amenities, productivity, and welfare serve as key drivers of migration flows under the *hukou* regime.

5.2 Mobility Cost and *Hukou*

The model estimates a comprehensive structure of bilateral migration frictions that includes both geographic and institutional components. These compound costs are represented as a 313×313 matrix of origin–destination migration barriers. Figure 9 presents the implied total cost of moving into each prefecture. To provide a meaningful comparison, I take the average of this cost conditional on destination prefectures (migrant recipient cities). In other words, the bilateral cost is expressed as a moving-in cost specific to each location. These costs were normalized relative to the cost in Shanghai.

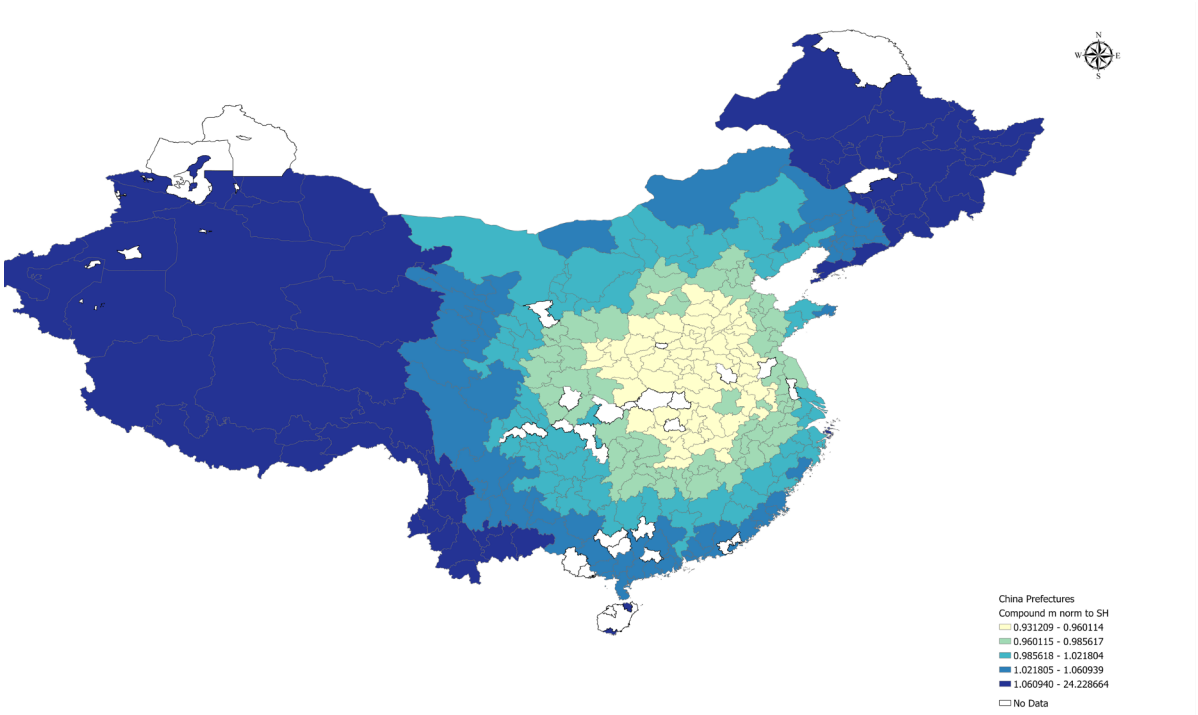


Figure 9: **Moving-in Cost at $t = 0$.**

The resulting map reveals a complex spatial pattern in which migration costs are influenced by both physical distance and infrastructural connectivity, alongside the institutional barriers created by the *hukou* system. While geographic isolation and poor infrastructure elevate migration costs in remote areas, even well-connected urban centers can impose high institutional barriers, making it difficult for migrants to access local services and benefits without the proper *hukou* registration.

To better understand how institutional frictions contribute to internal migration barriers, we calculate the *hukou* cost share for each city as the ratio of *hukou*-induced cost to the total inward migration cost (Figure 10). This component captures the relative importance of *hukou* restrictions—compared with physical distance, or geographic isolation—in shaping individuals' effective cost of relocating across prefectures.

The resulting distribution is highly uneven across space, with clear geographic and administrative patterns. In some cities, such as Suzhou, Lu'an, and Anqing in Anhui province, Luzhou, Neijiang in Sichuan province, and Longnan, Zhangye in Gansu province, the *hukou* share of total migration cost exceeds 75%, indicating that institutional restrictions—rather than geographic or economic factors—constitute the primary deterrent to migration. These cities typically feature rigid registration systems and limited integration pathways for migrants, amplifying the disutility associated with moving in.

In contrast, other cities such as Haikou (21%), Chengdu (27%), Zhoushan (38%), Chongqing (40%), Sanya (48%), and Wenzhou (49%) show low *hukou* cost shares, despite varying levels of compound migration cost. These cases reflect environments where either institutional policies are more flexible or physical remoteness, trade frictions, or low connectivity dominate the migration friction landscape.

This variation across the two maps shows that some cities are dominated by institutional barriers, while others are more affected by connectivity and geographic isolation. Thus, the interplay between physical and institutional frictions is highly location-specific.

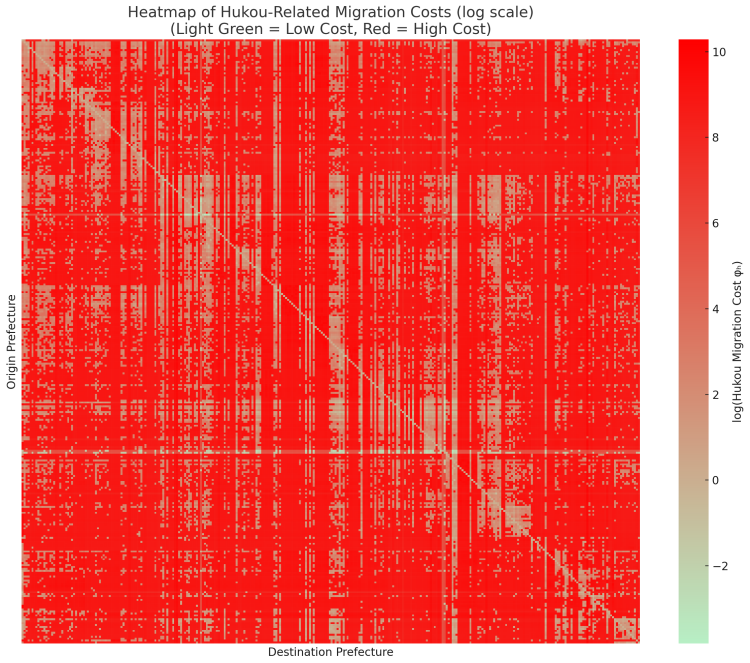


Figure 10: ***Hukou* Induced Migration Cost (log-scale).**

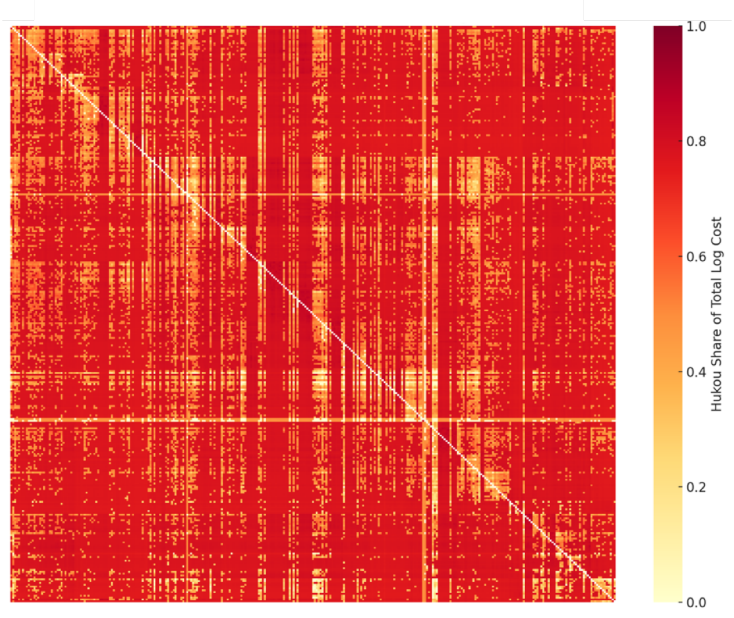


Figure 11: **Share of *Hukou* Cost in Total Migration Cost (log-scale).**

Many economically advanced cities—including Tier 1 cities such as Shenzhen (66%), Shanghai (60%), and Beijing (54%) fall into the middle range of the *hukou* cost share distribution. While these cities had not yet implemented any significant *hukou* policy reforms in 2000, they remained attractive migration destinations due to their high wages and superior amenities. The *hukou* barrier in these cases represents a meaningful share of total migration cost, but migrants were still willing to absorb these costs to access economic opportunities. Similarly, cities like Nanjing (53%) and Changsha (53%) show moderate *hukou* frictions, suggesting that institutional barriers were present but not necessarily the dominant constraint in shaping migration decisions.

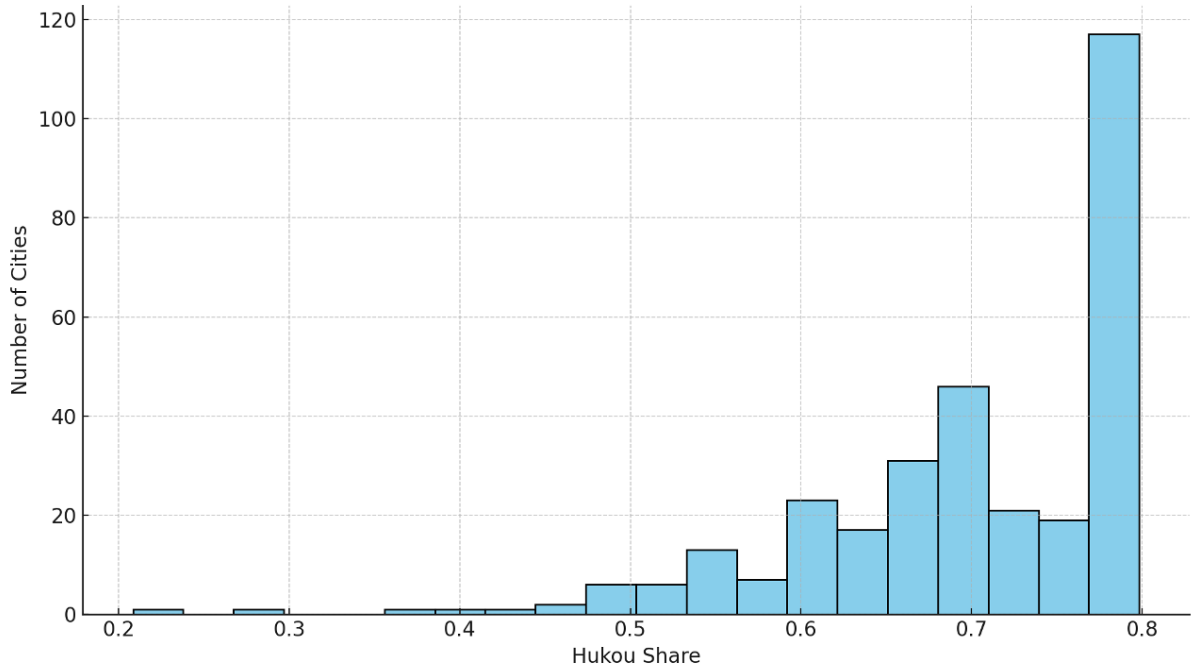


Figure 12: **Distribution of *Hukou* Cost Share in Total Migration Cost.**

Importantly, the data challenge common narratives that associate high *hukou* barriers solely with large cities. Despite perceptions that cities like Beijing and Shanghai are among the most difficult for migrants to enter, the model suggests that their high baseline utility and economic opportunities may offset *hukou* penalties in equilibrium. The *hukou* cost share is not solely a function of city size or economic development. Instead, it reflects a combination of local administrative practices, social service availability, and the structure of total migration costs. In some cases, migrants may perceive restrictive *hukou* policies as a larger burden than physical movement, while in others, distance and trade costs dominate the migration decisions.

This heterogeneity implies that reforming the *hukou* system could have differential impacts. In high-barrier cities, relaxing *hukou* restrictions could significantly lower total migration costs and promote labor reallocation. In contrast, in cities where *hukou* is already a minor factor, improvements in transport, housing, or wage opportunities may be more critical.

5.3 Model Validation

To assess the credibility and empirical relevance of the model, we conduct a validation exercise using untargeted moments. Specifically, we evaluate the model's ability to reproduce (i) provincial contributions to real GDP, (ii) population density across regions, and (iii) spatial patterns of labor mobility over time. These dimensions are critical for assessing whether the model reliably captures the dynamics of China's regional economy and internal migration during a period of rapid urbanization and structural transformation.

Figure 13 compares simulated and observed provincial GDP shares in the year 2000. The model successfully replicates broad spatial patterns in economic output, capturing the dominance of coastal provinces such as Guangdong, Jiangsu, and Zhejiang. These regions, historically prioritized for industrial development and foreign trade, contribute substantially to national GDP, both in the model and in the data. The model also reflects the growing economic role of inland provinces like Henan and Hubei, which have emerged as transportation

and manufacturing hubs in recent decades.

Despite this alignment, the model overestimates GDP shares for some leading provinces such as Guangdong and Shanghai, while underestimating Beijing's economic output. These discrepancies may stem from the model's assumption of static productivity in the initial period, which does not fully capture city-specific innovation or external shocks. Nonetheless, the overall fit is strong: the correlation coefficient between estimated and observed GDP shares is 0.94, indicating that the model provides a reasonable approximation of cross-provincial economic structure.

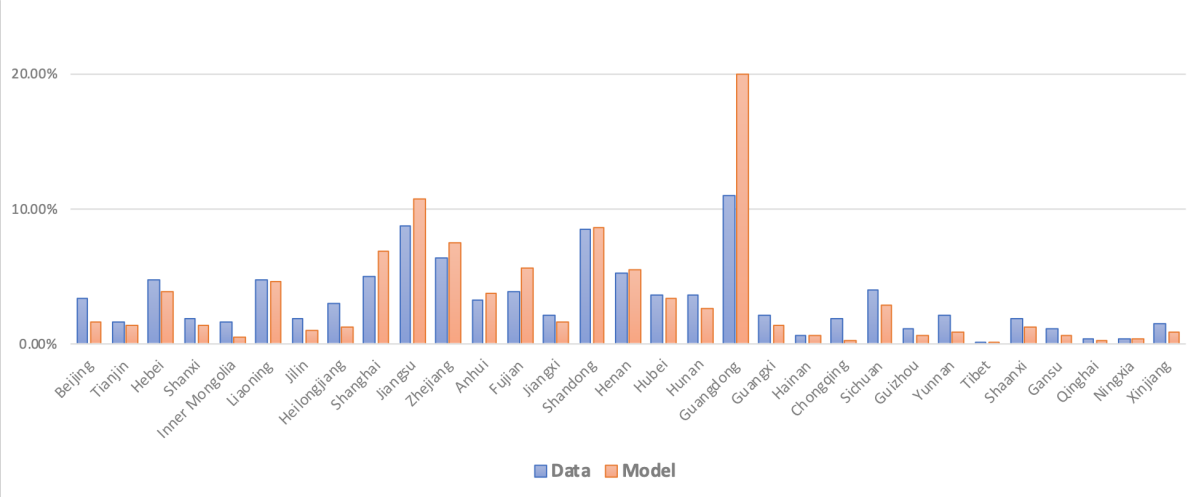


Figure 13: **Model VS. Data: Output Share in 2000.**

Next, we examine the model's performance in replicating spatial population patterns. Figure 13 shows scatter plots comparing predicted and actual population density (log-transformed) for 313 prefectures in 2010 and 2020. The model captures the broad demographic reallocation from inland to coastal regions, particularly the rapid urbanization of the Yangtze River Delta, the Pearl River Delta, and the Beijing-Tianjin-Hebei corridor. These regions attract large inflows of labor due to their superior productivity, infrastructure, and amenities—patterns that are well reflected in the simulated population distribution.

The model also reproduces labor outflows from interior provinces such as Gansu, Guizhou, and parts of Sichuan, where lower productivity and higher mobility frictions reduce local economic attractiveness. Some discrepancies remain, notably in high-density cities like Shanghai and Beijing, where the model slightly underpredicts population size. These deviations may result from not fully modeling land-use constraints, housing prices, or formal migration quotas, which could influence congestion and settlement patterns.

Overall, the model generates a strong empirical fit along both economic and demographic dimensions. It captures the essential drivers of regional development and labor mobility: spatial productivity gaps, endogenous amenities, trade access, and institutional barriers such as the *hukou* system. While some simplifying assumptions may understate second-order frictions, the observed alignment with untargeted moments supports the model's utility for policy simulations and counterfactual analysis in later sections.

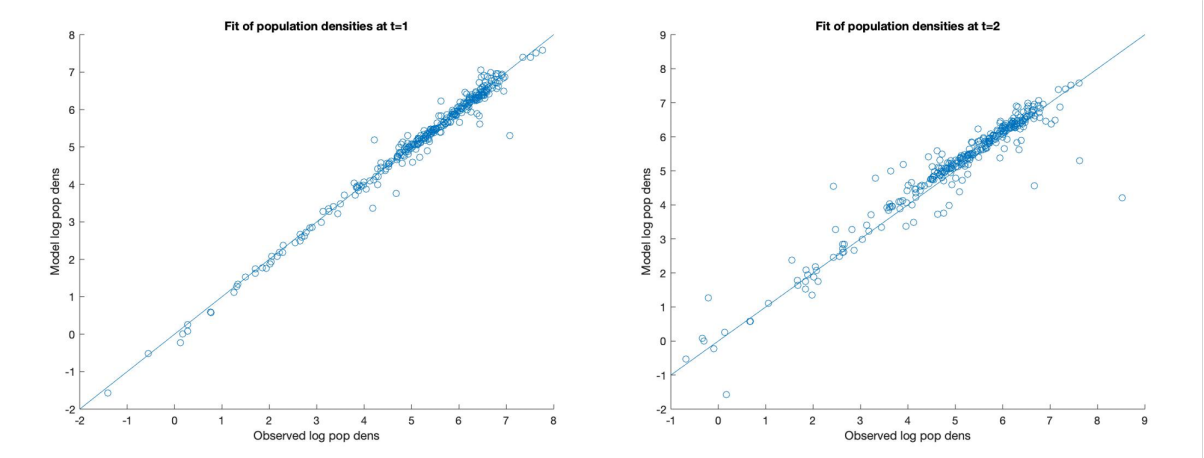


Figure 14: **Model VS. Data: Population Density in 2010 and 2020.**

6 Counterfactual Analysis: *Hukou* Reform Scenarios

In this section, we evaluate the economic and spatial implications of abolishing or partially removing the *hukou* system under different policy designs. We first consider three reform scenarios with varying geographic scope, then decompose the resulting welfare changes into their underlying mechanisms. We further examine the robustness of these results to alternative parameter values.

6.1 Reform Scenarios

We analyze the following policy scenarios, all relative to the baseline calibration with observed *hukou* frictions:

1. **S0 – Full abolition.** All *hukou*-related migration costs are removed nationwide, i.e., $\phi(h, n) = 1$ for all origin–destination pairs (h, n). Migration is subject only to physical or geographic costs $m(h, n)$.

2. **S1 – High-barrier cities only.** *hukou* restrictions are abolished only in destinations where the *hukou* cost share $\Phi(n) \geq 0.75$, covering 128 prefecture-level cities and roughly 36% of the national population. These are disproportionately inland or resource-dependent (e.g., mining) cities with weaker productivity fundamentals. For all other destinations, $\phi(h, n)$ remains at its baseline level.

3. **S2 – Economic hubs only.** *hukou* restrictions are abolished only in the top 10% of cities by baseline GDP share ($\geq 1\%$ in year 2000), covering 46 cities and about 18% of the population. These hubs are predominantly coastal or highly urbanized and benefit from strong agglomeration economies.

Figure 15 shows cities affected by different reform scenarios.

6.2 Aggregate Results

To compute the aggregate effects reported in Table 2, we use the following definitions. All changes are expressed relative to the baseline (no reform) economy.

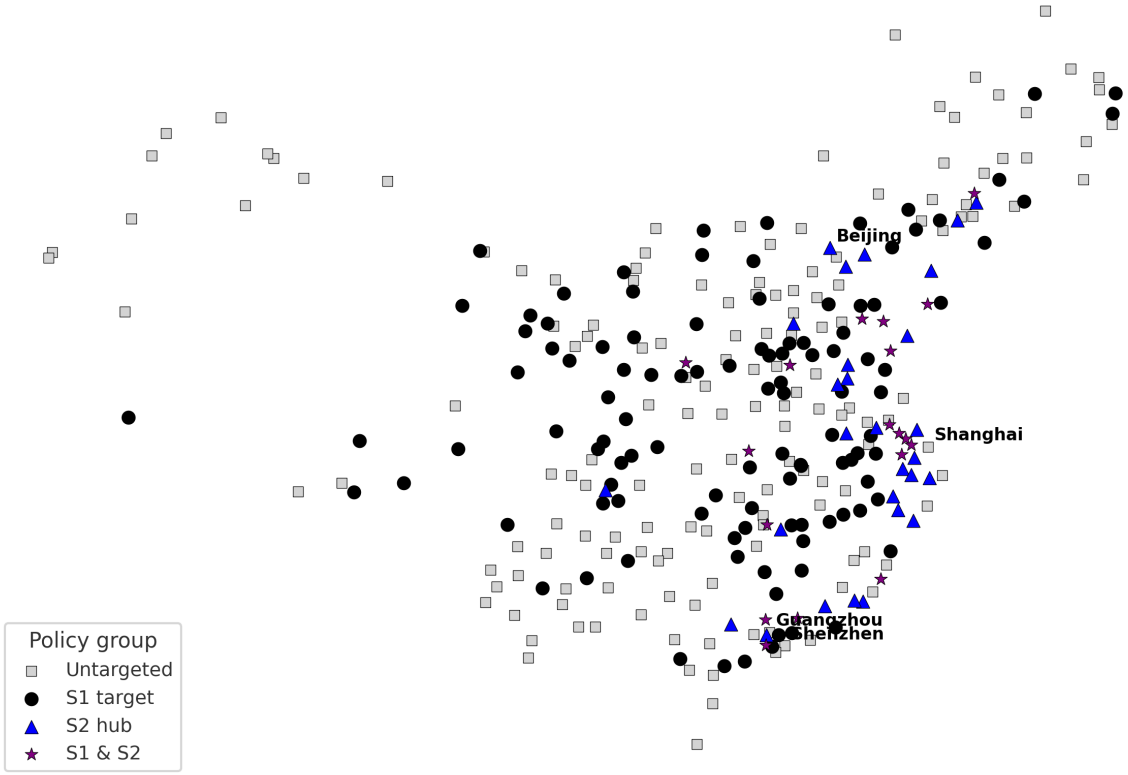


Figure 15: **Targeted Cities Under Different Reform Scenarios.**

Aggregate productivity is the summation of regional productivities:

$$\bar{Z}_t \equiv \sum_n Z_t(n). \tag{32}$$

The cumulative percentage change is then:

$$\Delta Z_t = 100 \times \left(\frac{\bar{Z}_t^{\text{CF}}}{\bar{Z}_t^{\text{Baseline}}} - 1 \right). \tag{33}$$

National output is the sum of prefecture-level outputs:

$$Y_t = \sum_n Y_t(n). \tag{34}$$

The reported change is:

$$\Delta Y_t = 100 \times \left(\frac{Y_t^{\text{CF}}}{Y_t^{\text{Baseline}}} - 1 \right). \tag{35}$$

Aggregate welfare is based on the population-weighted utility index:

$$U_t = \sum_n L_t(n) u_t(n), \tag{36}$$

where $u_t(n)$ is deterministic utility.

Define the consumption-equivalent factor ζ that makes the baseline path indifferent to scenario s under CRRA ($\rho = 2$) and discount factor β:

$$\zeta = \left[\frac{\sum_{t=0}^{T} \beta^t \left(U_t^s\right)^{1-\rho}}{\sum_{t=0}^{T} \beta^t \left(U_t^{\text{Baseline}}\right)^{1-\rho}} \right]^{\frac{1}{1-\rho}}. \tag{37}$$

We report the level-equivalent percentage change:

$$\Delta CE(T) \;=\; 100\,(\zeta - 1), \qquad T \in \{2, 10\}. \tag{38}$$

We do not annualize CE (see table notes) since with $\rho > 1$, annualizing can be misleading.

For each scenario s and period t, we define the migration rate as the share of the total population that has relocated relative to the initial distribution:

$$M_t^s \;=\; \frac{\text{Migrants}_t^s}{\text{Population}_t}, \tag{39}$$

where Migrants_t^s denotes the total number of individuals who change location in scenario s at time t, and Population_t is the aggregate population, which is constant across scenarios.

To evaluate the effect of *hukou* reform, we compute the difference in migration rates between scenario s and the baseline:

$$\Delta M_t^s \;=\; M_t^s \;-\; M_t^{\text{Baseline}}. \tag{40}$$

Because the denominator is identical across scenarios, ΔM_t^s captures the additional share of the population that migrates as a direct consequence of *hukou* reform. This metric is reported in the final column of Table 2.

Table 2 reports short-run ($t = 2$, 20 years) and long-run ($t = 10$, 100 years) percentage changes in aggregate productivity, real output, CE welfare, and migration rates across scenarios. CE welfare is computed as a level-equivalent measure following Equations (37)–(38), which capture the uniform increase in consumption under the baseline that would make households indifferent to the counterfactual. Because $\rho > 1$ makes annualization misleading, we report CE changes only at the $t = 2$ and $t = 10$ horizons.

Table 2: **Aggregate Effects of *Hukou* Reform Scenarios.**

	Productivity		Real Output		CE Welfare		Migration
Scenario	$t = 2$	$t = 10$	$t = 2$	$t = 10$	$t = 2$	$t = 10$	$t = 2$
Cumulative change (%)							
S0: Full abolition	-7.90	698.39	-3.55	972.06	48.51	52.89	83%
S1: High-barrier only	-6.75	38.30	-8.66	37.71	-0.37	0.17	72%
S2: Economic hubs only	2.86	199.96	19.11	169.43	0.59	1.36	64%
Annualized growth (%/yr) for $t = 10$ *(geometric)*							
S0: Full abolition	–	2.10	–	2.89	–	–	
S1: High-barrier only	–	0.34	–	0.33	–	–	
S2: Economic hubs only	–	1.13	–	1.00	–	–	

Notes: $t = 2$ and $t = 10$ correspond to 20 and 100 years after reform, respectively. Annualized growth rates are geometric: $g_X = \big(1 + \Delta_X/100\big)^{1/100} - 1$, computed from the cumulative changes shown above. CE welfare is reported only as a cumulative level-equivalent percentage change (not annualized). Because $\rho > 1$ in the CRRA mapping, directly annualizing CE can be misleading (sign-reversing). Migration is reported as the share of the population that changes prefecture by $t = 2$.

Removing *hukou* barriers nationwide triggers a large initial migration surge, with many movers drawn to high-amenity but only moderately productive destinations. This compositional shift causes short-run declines in aggregate productivity (–7.9%) and output (–3.55%) as measured at $t = 2$. Over time, however, agglomeration effects in high-productivity hubs

and endogenous amenity investments raise efficiency across the system, reversing the initial dip and producing dramatic long-run gains. After a century ($t = 10$), productivity and real output stand 699% and 972% above baseline, corresponding to annualized growth rates of 2.10%/yr and 2.89%/yr. CE welfare rises sharply—+48.5% in the short run and +52.9% in the long run—driven by both direct reallocation gains and dynamic feedbacks from productivity and amenity growth. The trajectory is consistent with the divergence patterns documented in Section 6.3: while average productivity growth initially slows due to inflows into mid-tier cities, over the long run the most productive hubs expand their lead, accounting for much of the aggregate gain.

Scenario S1 (reforms in high-barrier cities only) redirects labor toward many moderate-productivity inland cities, producing short-run declines in productivity (–6.75%) and output (–8.66%) and slightly negative CE welfare (–0.37%). By $t = 10$, productivity and output are 38.3% and 37.7% above baseline, but growth rates remain modest (0.34%/yr and 0.33%/yr). CE welfare is essentially unchanged in the long run (+0.17%), as efficiency gains are small and congestion costs limit amenity improvements.

Scenario S2 (reforms in top economic hubs only) liberalizes migration into 46 top-GDP cities, generating immediate productivity (+2.86%) and output (+19.11%) gains. By $t = 10$, productivity and output are 200% and 169% above baseline (1.13%/yr and 1.00%/yr). However, long-run CE welfare rises only 1.36%. Migration is lower than in S0 (64%), consistent with the more restricted set of destinations.

Taken together, the results show that full liberalization produces the largest and most persistent aggregate benefits, while partial reforms yield smaller or negligible welfare improvements. The CE metric highlights that efficiency gains do not translate one-for-one into welfare once congestion and cost-of-living effects are incorporated.

6.3 Growth Dynamics

Figure 16 compares the growth rates of aggregate productivity, real GDP, and utility across the Baseline, S0, S1, and S2 scenarios over the ten decades of the simulation.

6.3.1 Real GDP Growth

Nationwide removal of *hukou* restrictions (S0) produces the largest and most persistent increase in real GDP growth relative to the Baseline, reflecting substantial reallocation of labor toward more productive locations. Targeted reforms also raise GDP growth but to a lesser extent: S1 covers more prefectures (128), though many are only moderately productive, while S2 focuses on 46 top economic hubs, generating more concentrated gains despite covering fewer cities.

6.3.2 Productivity Growth

Across reform scenarios, average productivity growth rates fall slightly below the Baseline. This does not imply convergence. Instead, it reflects the concavity of agglomeration effects in the model. As major hubs absorb additional migrants, the marginal productivity gains from further concentration diminish, lowering the national average growth rate even as divergence between hubs and the periphery continues.

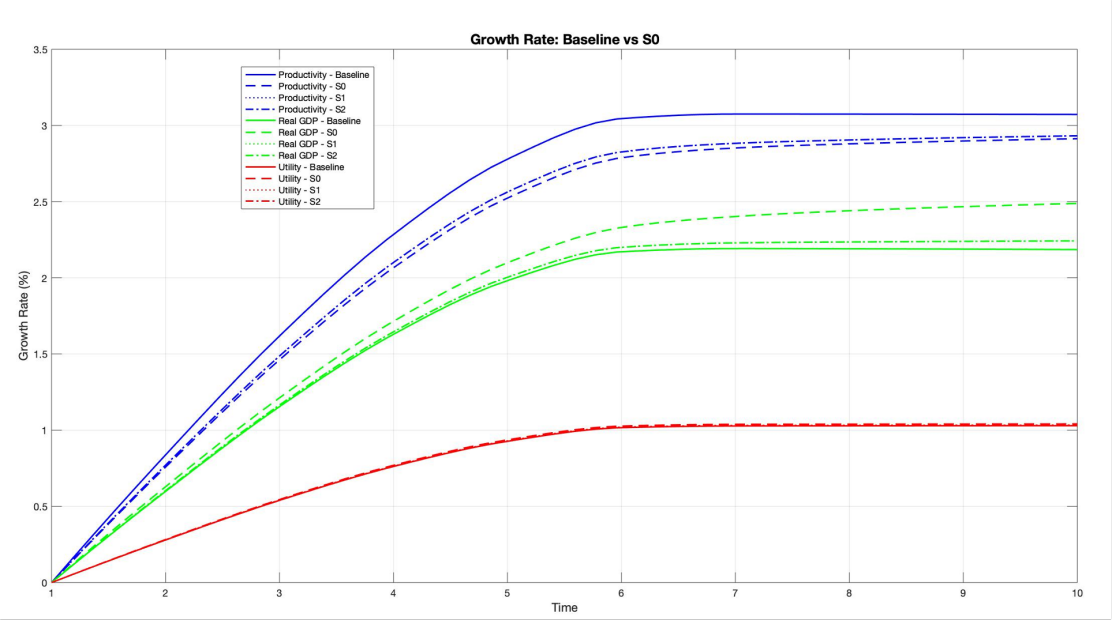


Figure 16: **Growth Dynamics.**

6.3.3 Utility Growth

Utility growth rates look nearly indistinguishable across scenarios. This apparent similarity is due to the offsetting effect of higher living costs and amenity congestion in the expanding hubs, which erode much of the wage gains from reallocation. However, the long-run consumption-equivalent (CE) welfare diverges. In S0, the economy experiences an immediate and substantial upward shift in utility levels, leading to a 52.89% higher CE welfare after 100 years compared to the Baseline, even though the long-run growth rate of utility is similar across scenarios. S1 and S2, by contrast, generate smaller and later level shifts in utility, resulting in much smaller CE gains.

Overall, these results highlight that the scope and targeting of migration reforms matter. Broad national liberalization produces the largest GDP and welfare gains, while targeted reforms yield more modest improvements. At the same time, GDP growth gains do not necessarily translate into higher long-run utility growth rates, underscoring the importance of considering both the level effects and general equilibrium feedbacks—such as land rents and congestion—when evaluating the welfare implications of migration policy.

6.4 Spatial Inequality Dynamics

Figures 17–19 reveal distinct patterns in the evolution of spatial inequality across wages, productivity, and utility.

6.4.1 Wage Inequality

In the Baseline, the Gini coefficient rises from 0.27 to 0.31 over ten periods, and the P90–P10 wage gap grows by roughly 15%. With full abolition (S0), the Gini climbs from 0.26 to 0.38, and the P90–P10 gap more than doubles, reflecting rapid migration toward high-wage cities. Targeting economic hubs (S2) produces the steepest divergence: the Gini exceeds 0.40 and the wage gap expands by nearly 150% relative to period 1. By contrast, the high-barrier-only reform (S1) yields an intermediate path, with inequality increases smaller than in S0 or S2.

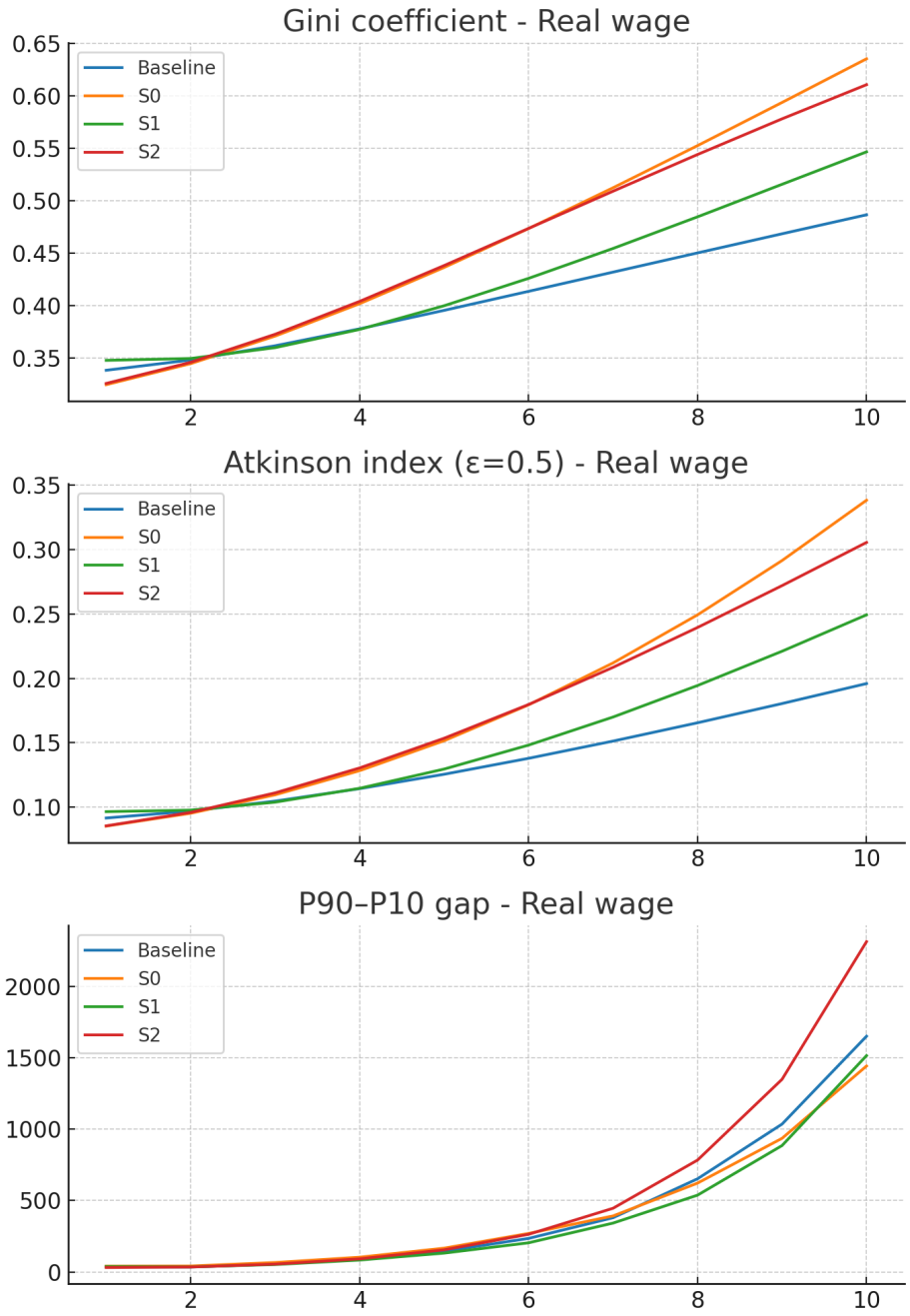


Figure 17: **Divergence or Convergence: Real Wage Inequality.**

6.4.2 Productivity Inequality

The spatial dispersion of productivity exhibits patterns similar to those for wages but with even stronger divergence forces. As shown in Figure 18, the Baseline already displays gradual widening over time, with the standard deviation of log productivity rising from 0.61 to 1.21 over ten periods. Under complete *hukou* removal (S0), dispersion nearly doubles over the horizon, while targeted reform toward advantaged cities (S2) produces the steepest trajectory, reaching 1.45 by period 10. These results indicate that migration-induced agglomeration effects compound pre-existing productivity advantages, with the largest gains accruing to cities already well-positioned in the productivity distribution. The similarity of patterns in wage and productivity dispersion suggests that real wage divergence is driven largely by underlying productivity

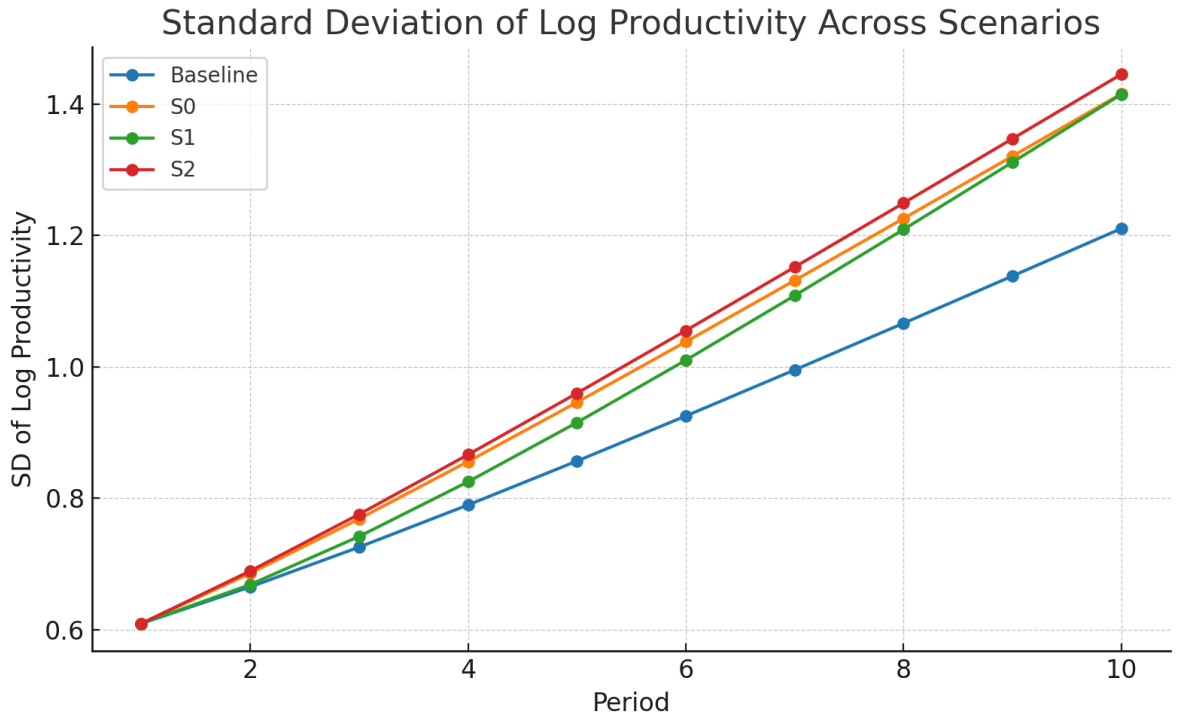


Figure 18: **Divergence or Convergence: Productivity Inequality.**

dynamics rather than wage-setting frictions.

6.4.3 Utility Inequality

In contrast, the dispersion of log utility tells a different story (Figure 19). Across all scenarios, the standard deviation of log utility either declines or remains nearly flat over time. For example, in S0, dispersion falls slightly from 0.042 in period 1 to 0.033 by period 10, and in S2, the decline is of similar magnitude. This muted or negative trend, despite strong divergence in wages and productivity, can be explained by congestion and amenity effects in top cities. While migrants moving to high-wage, high-productivity cities enjoy higher incomes, these gains are partially offset by higher costs of living, reduced amenities, and crowding effects. As a result, the spatial distribution of welfare is more equal than the distributions of wages or productivity alone would suggest.

Overall, the model predicts that while *hukou* reforms—especially those focused on top-tier cities—can sharply increase income and productivity dispersion, welfare inequality remains contained due to offsetting forces. This highlights a tension: visible economic divergence coexists with muted welfare divergence, a point relevant to policy design.

6.5 Spatial Patterns of Adjustment

Based on the results in Table3, the abolition of the *hukou* system under Scenario S0 generates highly uneven spatial adjustments in both labor flows and welfare outcomes by $t = 10$.

On the migration side, the largest net inflow gains are concentrated in mid-sized inland and lower-tier coastal cities rather than in the megacities. Tongling, Ezhou, Ma'anshan, and Huainan—largely clustered in Anhui and surrounding provinces—each gain over 350,000 residents (measured in 10,000s). Several resource-based or industrial cities, such as Hebi, Huaibei,

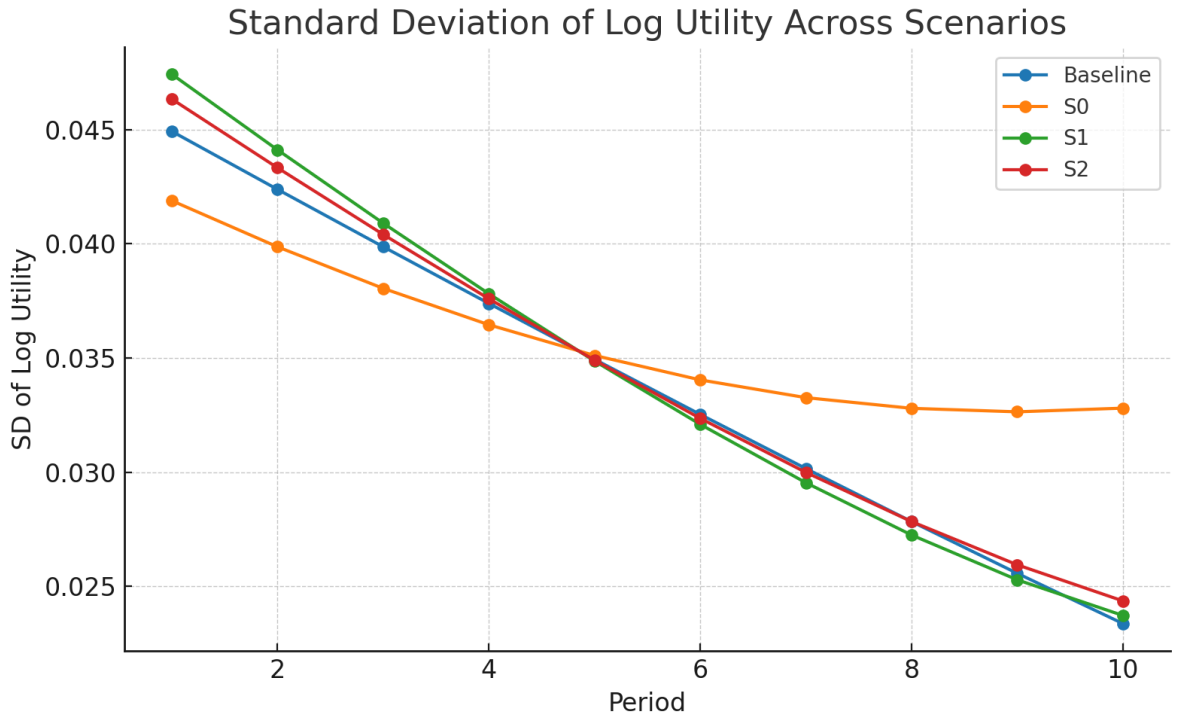


Figure 19: **Divergence or Convergence: Utility Inequality.**

and Wuhai, also register substantial inflows. This pattern suggests that once institutional barriers are lifted, labor reallocates not only toward high-wage coastal hubs but also toward medium-sized industrial centers with underutilized productive capacity.

The welfare results, however, tell a somewhat different story. The top CE welfare gains are dominated by cities starting from a relatively low baseline utility level. Jiayuguan, Huaiyin, and Anshun record increases exceeding 250% in consumption-equivalent welfare, reflecting large relative improvements from a low base. In several cases, such as Hebi and Luohe, cities appear in both the top inflow and top welfare gain lists—indicating that their gains are driven by both increased scale and improved economic conditions.

Conversely, the largest net outflows are concentrated in remote or less-connected regions, such as Ku'erle, Delingha, and Alxa, many located in Xinjiang, Qinghai, and Inner Mongolia. These regions face outmigration exceeding half a million people, reflecting both their geographic isolation and lower productivity in the counterfactual equilibrium.

In terms of welfare losses, the bottom-ranked cities are predominantly in either highly touristic coastal regions (e.g., Haikou, Sanya, Zhoushan) or large metropolitan areas such as Chengdu, Chongqing, and Wuhan. In tourist destinations, welfare declines are likely driven by congestion and increased costs without commensurate productivity gains. In large metros, the loss may be partly due to amenity dilution from rapid inflows exceeding infrastructure capacity, offsetting the benefits from scale economies.

Overall, these results highlight that removing *hukou* constraints creates a clear divergence between migration winners and welfare winners. Cities that attract the most migrants are not always those experiencing the largest proportional welfare gains, and some high-productivity urban centers may even see short-run welfare declines if congestion effects dominate. This underscores the importance of complementary urban policies on infrastructure, housing, and public services during large-scale migration adjustments.

Table 3: **Largest Movers and Welfare Changes Under Full Abolition.**

Top 10 net inflow (t=10)				Top 10 welfare gain (t=10, CE %)		
City	$\Delta\bar{l}$	Rank	Hukou cost (%)	City	CE %	Rank
Tongling	+43	1	75	Jiayuguan	394	1
Ezhou	+42	2	79	Huaiyin	354	2
Ma'anshan	+38	3	79	Anshun	257	3
Huaiyin	+37	4	80	Liangyungang	218	4
Hebi	+37	4	79	Hebi	174	5
Luohe	+35	6	79	Hunjiang	168	6
Huaibei	+34	7	70	Luohe	162	7
Heyuan	+34	8	78	Jicheng	158	8
Wuhai	+32	9	74	Heze	147	9
Xiamen	+31	10	70	Tongling	146	10

Bottom 10 net inflow ($t = 10$)			Bottom 10 welfare gain (t=10, CE %)		
City	$\Delta\bar{l}$	Rank	City	CE %	Rank
Ku'erle	-61	1	Haikou	-100	1
Delingha	-59	2	Sanya	-99	2
Alxa	-56	3	Zhoushan	-99	3
Naggu	-53	4	Jingdezhen	-90	4
Hailar	-53	4	Chengdu	-89	5
Xilingguole	-52	6	Chongqing	-85	6
Yushu	-50	7	Wuhan	-78	7
Yining	-49	8	Tongchuan	-70	8
Jiuquan	-46	9	Fuzhou	-59	9
Hetian	-44	10	Jiaxing	-58	10

Notes: Net inflow measured in units of 10,000 people. CE welfare computed with $\rho = 2$.

Figures 20 show long-run spatial changes in net labor inflows (top panels), and deterministic welfare (bottom panels) for three counterfactual scenarios at $t = 10$. In each map, the ten largest and ten smallest cities in each metric are highlighted, illustrating the highly uneven spatial responses.

In a full abolition scenario, migration inflows are concentrated in mid-sized industrial and inland cities in Anhui, Henan, and Inner Mongolia, alongside some smaller coastal manufacturing hubs. Tongling (+430 k), Ezhou (+420 k), and Ma'anshan (+380 k) are among the top destinations, reflecting underutilized productive capacity. Outflows dominate in remote western prefectures such as Ku'erle (–610 k) and Delingha (–590 k), and in certain large metropolitan areas where congestion effects offset wage advantages. Welfare gains are largest in lower-baseline inland cities—Jiayuguan (+394%), Huaiyin (+354%), and Anshun (+257%), while tourist-dependent cities like Sanya (–99%) and high-density metros such as Chengdu (–89%) register losses.

In a high-barrier-only scenario, *hukou* restriction is lifted in 128 high-barrier cities (*hukou* cost share $\geq$ 75%), many of which are inland or resource-dependent. Net inflows are led by Hebi (+370 k) and Luohe (+350 k), but these destinations tend to have low initial productivity

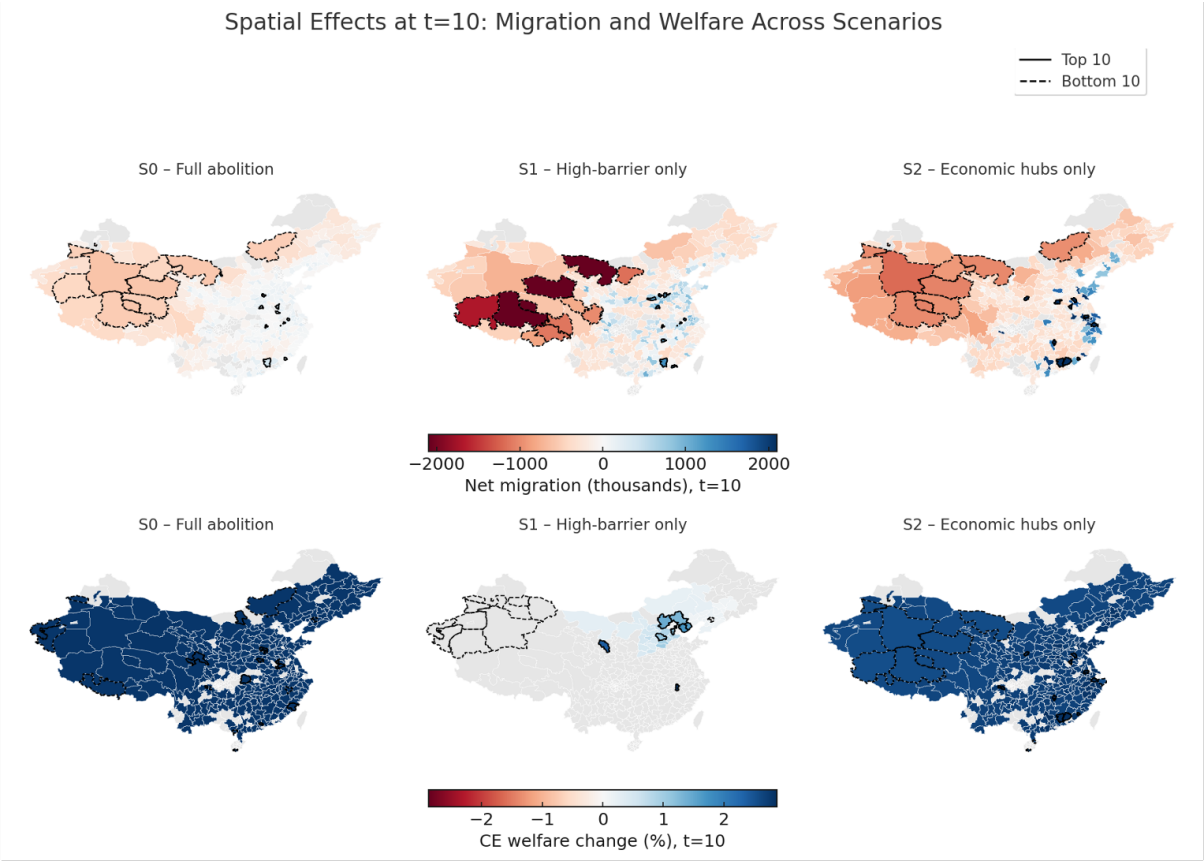


Figure 20: **Net Labor Inflow and Welfare in the Long-Run.**

and limited agglomeration economies, driving short-run declines in aggregate productivity and welfare. Gains are localized—Hunjiang (+168%) and Jicheng (+158%)—while large swathes of the coast and major hubs remain unaffected.

If *hukou* is lifted in economic hubs only, it means liberalizing migration into 46 top-GDP cities (each contributing $\geq 1\%$ of baseline GDP) channels inflows toward Beijing, Shanghai, Shenzhen, and key provincial capitals. Short-run productivity and output rise due to scale economies, but welfare gains are uneven. Some secondary hubs in the liberalized group see modest improvements—e.g., certain Pearl River Delta cities record CE welfare gains of about +2.8%—yet several megacities face welfare declines from congestion and amenity dilution.

Across all scenarios, migration winners and welfare winners do not perfectly overlap. Cities attracting the largest inflows are not always those with the highest proportional welfare gains, and some high-productivity or high-density metros experience welfare losses from congestion and amenity dilution when inflows outpace the capacity of existing amenities and infrastructure in the model. The central implication is that migration liberalization alone produces uneven local outcomes, shaped by initial conditions and adjustment dynamics.

Productivity patterns broadly mirror, but do not perfectly align with the migration outcomes. In Scenario S0, productivity gains are broad-based, with the largest effects in coastal hubs and several inland poles. This pattern is consistent with agglomeration feedback operating on both productivity and wages once migration frictions are removed. In S1, many targeted inland cities show positive migration and wage responses but muted or even short-run declines in measured productivity. This reflects the dilution of average productivity when low-productivity workers move in faster than local technology or scale economies can adjust. Moreover, lo-

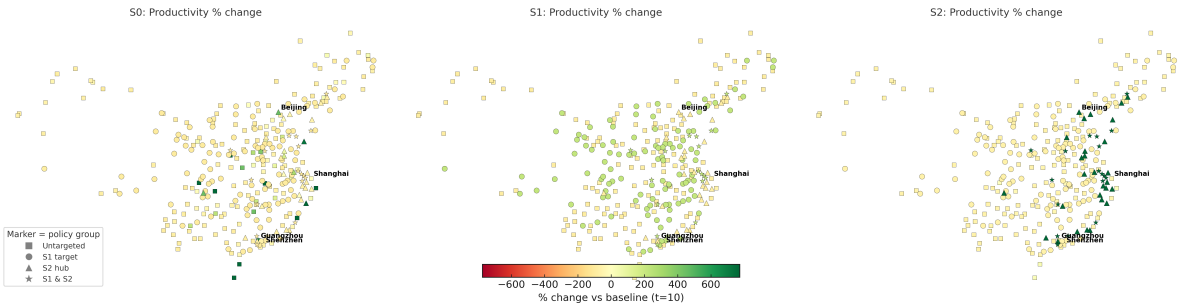


Figure 21: **Productivity Change at $t = 10$.**
Notes: Values are relative to the baseline. Colors indicate percent change. Markers denote policy groups: □ Untargeted, ∘ S1, △ S2, ⋆ S1&S2.

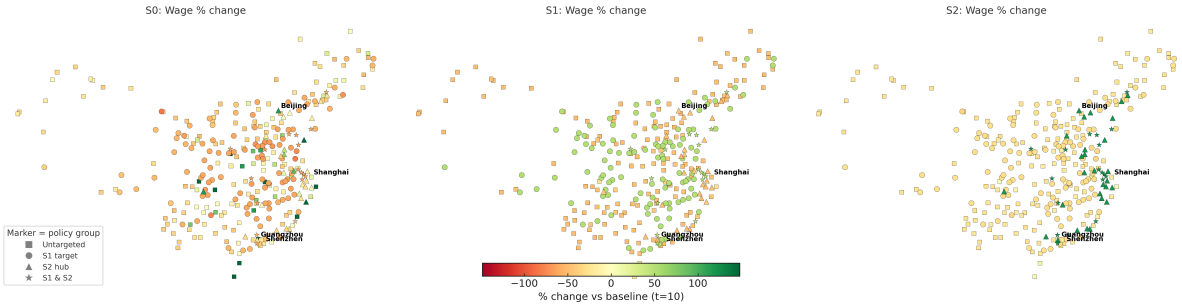


Figure 22: **Wage Change at $t = 10$.**
Notes: Values are relative to the baseline. Colors indicate percent change. Markers denote policy groups as in Fig. 21.

cal scale/technology adjusts with a lag. By contrast, S2's targeted reform in economic hubs delivers immediate productivity boosts in the short run. But over longer horizons, marginal agglomeration returns diminish and congestion/price pressures temper the gains.

Across scenarios, productivity and wage maps are positively related but not one-for-one: in dense recipients, prices rise faster than wages (via non-tradables/land), so real income gains are smaller than productivity gains; in outflow areas, both wages and productivity fall.

6.6 Mechanism Decomposition

The aggregate and spatial patterns documented above are the net outcome of multiple offsetting forces operating through the model's structural channels. To disentangle these forces, we decompose the total consumption-equivalent (CE) welfare gain into three structural components, plus their interaction:

1. **Pure reallocation.** Labor reallocates across space; productivity $\tau_t(n)$ and amenities $A_t(n)$ are fixed at baseline levels.

2. **Productivity feedback.** Allow $\tau_t(n)$ to update via the agglomeration elasticity λ_1, while holding amenities fixed.

3. **Amenity feedback.** Allow amenities to respond via λ and χ, while holding productivity fixed.

4. **Interaction.** The residual complementarities between productivity and amenity channels when both are active.

We compute the contribution of each channel relative to a common baseline in which only the *hukou* barrier is removed (migration-only scenario). That is, productivity and amenity feedbacks are each "added on" to the migration-only baseline rather than to one another. As a result, their magnitudes cannot be summed directly. The remaining gap between the sum of these single-channel increments and the full model outcome is reported as the interaction term, which captures complementarities when productivity and amenity channels operate simultaneously.

Under full *hukou* abolition, long-run welfare rises by about 53 pp in CE terms. Pure reallocation explains 18.6 pp (35.1% of the total), productivity feedback contributes 9.8 pp (18.5%), and amenity feedback adds 15.6 pp (29.4%). The remaining 8.9 pp (16.7%) reflects complementarities, indicating that productivity and amenity channels reinforce each other when both are active.

These magnitudes reflect two structural features of the Chinese economy. First, amenity-driven migration is a prominent channel: local governments often invest heavily in public services, infrastructure, and urban livability—financed in part by land revenues—which directly raises location attractiveness and accounts for over half of the welfare gain in the simulation. Second, migration rates are already high in many regions, particularly among young and rural-to-urban workers. As a result, removing the *hukou* barrier alone—while holding productivity and amenities fixed—delivers a sizeable welfare gain, largely through improved matching and reduced mobility costs. Nevertheless, the largest welfare improvements arise when dispersion from lower migration costs is coupled with agglomeration forces from endogenous productivity and amenity growth, producing mutually reinforcing adjustments that reshape China's spatial economy.

Table 4: **Mechanism Decomposition of Long-Run Welfare Gains at $t = 10$.**

Scenario	Incremental (pp)	Share of Total (%)
Migration only (ϕ only)	+18.60	35.2
Add-on effects (relative to ϕ only):		
ϕ + Productivity feedback (λ_1 on; A fixed)	**+9.80**	18.5
ϕ + Amenity feedback (λ, χ on; Z fixed)	**+15.57**	29.5
Interaction ($\phi \times Z \times A$)	**+8.9**	16.8
Full model($\phi + \lambda_1 + \lambda, \chi$)	**52.87**	**100.0**

Notes: All increments are measured relative to the migration-only (ϕ only) situation. The productivity and amenity effects are not additive; their combined operation generates an interaction effect. Together, the baseline + add-ons + interaction sum to the full model gain.

Finally, to ensure that the results are not driven by specific modeling choices or parameter values, I conduct a series of robustness checks, including alternative migration elasticity estimates, amenity specifications, and trade elasticities. The main findings are stable across these exercises. Detailed results are reported in Appendix H.

7 Conclusion and Discussion

This paper develops a dynamic spatial equilibrium model with endogenous productivity, amenities, and migration frictions to study how China's *hukou* system shapes labor allocation, re-

gional development, and long-run economic growth. The framework emphasizes three interacting mechanisms: spatial labor reallocation, agglomeration-driven productivity dynamics, and amenity investment financed through land rents. Together, these channels generate persistent feedback effects between migration, productivity growth, public investment, and local livability.

The quantitative analysis suggests that institutional migration barriers generate substantial long-run economic costs. By limiting migration toward more productive urban regions, *hukou*-related frictions weaken agglomeration effects and slow productivity growth over time. Under full reform, aggregate productivity and real output increase by 2.10% and 2.89% annually over the century, respectively, while long-run consumption-equivalent welfare rises by approximately 53%. A decomposition of the welfare gains attributes 35.1% to pure spatial reallocation, 18.5% to productivity feedbacks, and 29.4% to amenity feedbacks, with the remaining gains driven by interactions across channels.

The analysis also highlights the importance of local public-investment dynamics in shaping spatial development. In the model, economically dynamic regions generate higher land rents, which finance amenity-enhancing expenditures such as infrastructure and public services. As a result, migration restrictions influence not only labor allocation and productivity growth, but also the spatial distribution of fiscal capacity and urban livability. This mechanism is particularly relevant in the Chinese context, where local governments rely heavily on land-based revenues to support urban development.

The results further suggest that the effects of migration reform depend critically on its geographic scope and policy design. Reforms concentrated in major economic hubs generate larger short-run efficiency gains but also intensify congestion pressures and regional divergence over time. In contrast, broader reforms that reduce barriers across regions produce more balanced spatial outcomes, although with more moderate gains in aggregate efficiency. These findings illustrate the trade-off between maximizing aggregate growth and promoting spatially inclusive development.

An important implication of the model concerns the distinction between income inequality and welfare inequality. While relaxing migration restrictions tends to increase wage and productivity dispersion as agglomeration benefits accumulate in already-advantaged regions, utility inequality rises much less sharply because higher nominal income in large cities is partially offset by congestion costs and amenity crowding. This divergence highlights the importance of accounting for non-monetary dimensions of welfare when evaluating spatial and migration policies.

More broadly, the paper argues that migration institutions are not merely administrative arrangements governing labor mobility. They are fundamental components of development strategy that shape how workers, economic activity, and public-investment resources are distributed across space. In rapidly urbanizing economies, institutional mobility restrictions can therefore influence not only migration patterns, but also the long-run trajectory of productivity growth, structural transformation, and regional inequality.

These findings suggest that migration reform should not be viewed in isolation from urban planning, infrastructure provision, and regional development policy. Policies that expand labor mobility while simultaneously improving public-service access, investing in lagging regions, and managing congestion in highly concentrated cities are more likely to achieve both efficiency gains and inclusive development outcomes. More generally, the analysis highlights the importance of coordinated institutional and spatial policies in shaping sustainable long-run development.

Several limitations remain and point to directions for future research. The current framework abstracts from endogenous firm innovation and capital accumulation, focusing instead on

migration and agglomeration as the primary drivers of regional productivity dynamics. Future work could incorporate richer innovation processes, housing-market dynamics, local fiscal competition, and international trade linkages to further explore how migration institutions interact with broader processes of urbanization and economic development.

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Appendix A Derivations of Demand and Supply Equilibrium Conditions

This appendix provides detailed derivations of trade shares, price index, migration shares, initial period productivity, and exogenous amenities.

A.1 Derivation of the Trade Shares

Let $p_t^\omega(n,s)$ be the price of good ω produced in location n and purchased in location s at time t.

From the FOCs, we can have a closed form for $p_t^\omega(n,s)$:

$$p_t^\omega(n,n) = \iota^{-1} w_t(n) l_t(n)^{1-\iota} Z_t(n)^{-1} \tag{A.1}$$

where $Z_t(n) = Z_{t-1}(n)^\alpha \bar{l}_{t-1}(n)^{\lambda_1} \epsilon_t^\omega(n)$.

The price of good ω produced in location n and sold in location s is its locational price multiplied by the iceberg transportation cost.

$$\begin{aligned} p_t^\omega(n,s) &= \upsilon(n,s) p_t^\omega(n,n) \\ &= \upsilon(n,s) \iota^{-1} w_t(n) l_t(n)^{1-\iota} Z_t(n)^{-1}. \end{aligned} \tag{A.2}$$

The cumulative distribution function (CDF) of the prices for a good ω produced in location n and consumed in location s is:

$$\begin{aligned} &\mathbb{P}\left[p_t^\omega(n,s) \le p\right] \\ &\quad = \mathbb{P}\left[\upsilon(n,s)\iota^{-1} w_t(n) l_t(n)^{1-\iota} Z_t(n)^{-1} \le p\right] \\ &\quad = \mathbb{P}\left[\epsilon_t^\omega(n) \ge \frac{\upsilon(n,s)\iota^{-1} w_t(n) l_t(n)^{1-\iota} Z_t(n)^{-1}}{p}\right] \\ &\quad = 1 - \mathbb{P}\left[\epsilon_t^\omega(n) \le \frac{\upsilon(n,s)\iota^{-1} w_t(n) l_t(n)^{1-\iota} Z_t(n)^{-1}}{p}\right] \\ &\quad = 1 - exp\left\{-(\frac{\upsilon(n,s)\xi_t(n)}{p})^{-\delta}\right\}, \end{aligned} \tag{A.3}$$

where $\xi_t(n) = \upsilon(n,s)\iota^{-1} w_t(n) l_t(n)^{1-\iota} Z_t(n)^{-1}$

Then, the fraction of goods produced in location n and consumed in location s can be expressed as:

$$\begin{aligned} \pi_t(n,s) &= \mathbb{P}\left[p_t^\omega(n,s) \le \min_{j\in N} p_t^\omega(n,j)\right] \\ &= \int_0^\infty \mathbb{P}\left[\min_{j\in N} p_t^\omega(n,j) \ge p\right] dG_{ns,t}(p) \\ &= \int_0^\infty \prod_{j\in N} \mathbb{P}\left[p_t^\omega(n,j) \ge p\right] dG_{ns,t}(p) \\ &= \int_0^\infty \prod_{j\in N} \left[1 - G_{nj,t}(p)\right] dG_{ns,t}(p). \end{aligned} \tag{A.4}$$

If we replace $G_{ns,t}(p)$ by its expression in (19), we will have

$$\begin{aligned}\pi_t(n,s) &= \int_0^{\infty} \prod_{j\in N} exp\left\{(\frac{\upsilon(j,s)\xi_t(j)}{p})^{-\delta}\right\} exp\left\{(\frac{\upsilon(n,s)\xi_t(n)}{p})^{-\delta}\right\} (\upsilon(n,s)\xi_t(n))^{-\delta}\, dp^{\delta} \\ &= (\upsilon(n,s)\xi_t(n))^{-\delta} \int_0^{\infty} \exp\left\{-\sum_j^N (\upsilon(j,s)\xi_t(j))^{-\delta} p^{\delta}\right\} dp^{\delta} \\ &= \frac{(\upsilon(n,s)\xi_t(n))^{-\delta}}{\sum_j^N (\upsilon(j,s)\xi_t(j))^{-\delta}}\end{aligned} \tag{A.5}$$

This gives us the equation (13) in the text.

A.2 Derivation of Migration Flows

Let $\Omega_t(n,j)^h$ be the fraction of individuals, with *hukou* h, relocating from location j to location n at time t. That an individual chooses to relocate to n means she can maximize her utility in n at time t. So we have:

$$\begin{aligned}\Omega_t(n,j)^h &= \mathbb{P}\left[\tilde{u}_t(n)\tilde{m}_t(n,j)^{-1}\epsilon_t(n) \geq \tilde{u}_t(s)\tilde{m}_t(s,j)^{-1}\epsilon_t(s)\right] \\ &= \mathbb{P}\left[\tilde{u}_t(n)\tilde{m}_t(n,j)\epsilon_t(n) \geq \tilde{u}_t(s)\tilde{m}_t(s,j)\epsilon_t(s), s \neq n\right] \\ &= \mathbb{P}\left[\frac{\epsilon_{s,t}}{\epsilon_{n,t}} \leq \frac{\tilde{u}_t(n)^{-1}\tilde{m}_t(n,j)^{-1}}{\tilde{u}_t(s)\tilde{m}_t(s,j)}\right]\end{aligned} \tag{A.6}$$

Since $\epsilon_{n,t}$ follow a Fréchet distribution and i.i.d. across locations and times, the ratio of the them also follows a distribution with a CDF given by:

$$F(\epsilon \leq z) = \exp\left(-z^{-\gamma}\right)$$

The probability of choosing location n rather than location s then can be written as:

$$\mathbb{P}\left[\frac{\epsilon_t(s)}{\epsilon_t(n)} \leq \frac{\tilde{u}_t(n)\tilde{m}_t(n,j)}{\tilde{u}_t(s)\tilde{m}_t(s,j)}\right] = \exp\left(-\left(\frac{\tilde{u}_t(s)\tilde{m}_t(s,j)}{\tilde{u}_t(n)\tilde{m}_t(n,j)}\right)^{\gamma}\right)$$

The overall migration probability for location n is the probability that location n is chosen over all alternatives $s \neq n$. This the the product of the probabilities for each alternative location:

$$\begin{aligned}\mathbb{P} &= \prod_{s\neq n} \exp\left(-\left(\frac{\tilde{u}_t(s)\tilde{m}_t(s,j)}{\tilde{u}_t(n)\tilde{m}_t(n,j)}\right)^{\gamma}\right) \\ &= \exp\left(-\sum_{s\neq n}\left(\frac{\tilde{u}_t(s)\tilde{m}_t(n,s)}{\tilde{u}_t(n)\tilde{m}_t(n,j)}\right)^{\gamma}\right)\end{aligned} \tag{A.7}$$

Then the probability $\Omega_t^h(n,j)$ can be written as:

$$\Omega_t^h(n,j) = \frac{\tilde{u}_t(n)^{\gamma}\tilde{m}_t(n,j)^{-\gamma}}{\sum_s^N \tilde{u}_t(s)^{\gamma}\tilde{m}_t(n,s)^{-\gamma}} \tag{A.8}$$

Appendix B Initial Utility and Exogenous Amenity

The part of utility that does not depend on idiosyncratic amenity shocks, $\tilde{u}_t(s)$, is affected only by the characteristics of a location and is common for all its residents:

$$\tilde{u}_t(s) = \bar{A}_t(s)\bar{l}_t(s)^{\lambda}\frac{w_t(s)}{P_t(s)}$$

If we replace $P_t(s)$ by (18), we can have:

$$\tilde{u}_t(s) = \bar{A}_t(s)\bar{l}_t(s)^{\lambda} \frac{w_t(s)}{[\sum_s^N (\xi_t(n)v(n,s))^{-\delta}]^{-1/\delta} \left[\Gamma\left(\frac{1-\sigma}{\delta}+1\right)\right]^{\frac{1}{1-\sigma}}}$$

then, by rearranging the equation, we have:

$$\left[\Gamma\left(\frac{1-\sigma}{\delta}+1\right)\right]^{\frac{-\delta}{1-\sigma}} \sum_n^N (\xi_t(n)v(n,s))^{-\delta} = (\frac{1-\iota}{\iota})^{\chi} \left[\frac{\bar{A}_t(s)}{\tilde{u}_t(s)}\right]^{-\delta} \bar{l}_t(s)^{-\lambda\delta} w_t(s)^{-\delta(1+\chi)}. \quad \text{(B.1)}$$

Combining (A.5) and (B.1), the probability of goods produced in n and consumed in s becomes:

$$\pi_t(n,s) = (v(n,s)\xi_t(n))^{-\delta} \left[\Gamma\left(\frac{1-\sigma}{\delta}+1\right)\right]^{\frac{-\delta}{1-\sigma}} \left[\frac{\bar{A}_t(s)}{\tilde{u}_t(s)}\right]^{\delta} \bar{l}_t(s)^{\lambda\delta} w_t(s)^{\delta} \quad \text{(B.2)}$$

With (B.2), the trade balance condition can be expanded as:

$$w_t(n)H(n)\bar{l}_t(n)\xi_t(n)^{\delta} = \left[\Gamma\left(\frac{1-\sigma}{\delta}+1\right)\right]^{\frac{-\delta}{1-\sigma}} \sum_s^N \left[\frac{\bar{A}_t(s)}{\tilde{u}_t(s)}\right]^{\delta} H(s)\bar{l}_t(s)^{1+\lambda\delta} w_t(s)^{1+\delta} \upsilon_t(n,s)^{-\delta} \quad \text{(B.3)}$$

Plug in $\xi_t(n)$ and rearrange (B.3):

$$\begin{aligned} & w_t(n)^{1+\delta} H(n)\bar{l}_t(n)^{1+(1-\iota)\delta} \tilde{z}_t(n)^{-\delta} \\ = & \left[\Gamma\left(\frac{1-\sigma}{\delta}+1\right)\right]^{\frac{-\delta}{1-\sigma}} \sum_s^N \left[\frac{\bar{A}_t(s)}{\tilde{u}_t(s)}\right]^{\delta} H(s)\bar{l}_t(s)^{-\lambda\delta} w_t(s)^{-\delta(1+\chi)} \upsilon_t(n,s)^{-\delta} \end{aligned} \quad \text{(B.4)}$$

Appendix C Data Sources and Description

This appendix provides a detailed overview of the data sources and methodological approaches used in this study to calibrate the dynamic spatial equilibrium model and conduct the analysis of China's *hukou* system.

C.1 China Census Data

The primary data sources for this study include the China Census Data for the years 2000, 2010, and 2020, provided by the National Bureau of Statistics of China (NBS). The China Census is conducted every ten years and offers comprehensive demographic and socio-economic information across Chinese prefectures, including population size, age structure, education levels, employment status, and *hukou* status. These datasets were crucial for capturing changes in population distribution and labor mobility over time. The census data was accessed through the official NBS website, with some historical data obtained from academic institutions with special access to NBS archives.

C.2 City-level Statistical Yearbooks

Additional economic data was sourced from the City-Level Statistical Yearbooks, which are published annually by the NBS and various provincial and municipal statistics bureaus. These yearbooks provide detailed economic data for each prefecture-level city in China, including GDP per capita, total GDP, industrial output, disposable income, wages, and land area. This information was essential for understanding the economic context in which migration decisions are made and for calibrating the model to reflect regional economic disparities. The statistical yearbooks were accessed through the provincial statistics bureaus' online databases.

C.3 China Migrants Dynamic Survey (CMDS)

Conducted annually since 2010 by the National Health Commission of the People's Republic of China, it provided detailed information on China's migrant population. The CMDS data includes variables related to socio-economic status, health, education, employment, and migration history, which were integral to understanding the characteristics and behavior of the migrant population under the *hukou* system. The CMDS data was obtained through the official website of the CMDS, which requires specific authorization for years and regions.

C.4 Integrated Public Use Microdata Series (IPUMS)

Migration flow data from the initial period was derived from IPUMS International, which provides harmonized data from national censuses worldwide, including the 2000 China National Population Census. The dataset used in this study is a 1% sample of the total census population, offering a rich source of microdata on demographics, housing, employment, and migration. The IPUMS data was accessed through the IPUMS website, which provides public access to researchers upon registration.

In addition to demographic and economic data, climate variables such as average temperature, humidity, and sunlight hours were obtained from the China Meteorological Data Service Center, operated by the China Meteorological Administration. These variables were used as proxies for exogenous amenities in the model, influencing regional attractiveness. The climate data was accessed through the China Meteorological Data Service Center's official website, with some datasets requiring subscriptions.

Topographical and geographical data, including terrain relief and the presence of water bodies, were sourced from the National Geographic Information Center, affiliated with the Chinese Academy of Sciences. These data were used to measure exogenous amenities and their influence on population distribution. The Relief Degree of Land Surface dataset provided insights into the physical geography affecting regional development and settlement patterns. Data from the National Geographic Information Center was accessed through academic collaborations with the Chinese Academy of Sciences.

The ArcGIS OD Cost Matrix Tool, part of Esri's ArcGIS software suite, was employed to estimate bilateral trade costs between cities. This tool calculates the fastest paths through highways, roads, and railroads, allowing for the estimation of spatial frictions in the model by assessing the time and distance-based costs associated with transportation across regions. ArcGIS software was accessed through Esri's official website, with institutional licenses provided by Georgetown.

Appendix D Data Preparation

The methodology involved a rigorous calibration process, ensuring that the model accurately reflects the economic and demographic realities of Chinese prefectures from 1990 to 2020. Data processing and harmonization were critical to this effort. For census data, raw data from the NBS were processed to extract relevant variables such as population size, *hukou* status, and employment. This data was harmonized across different years to ensure consistency in definitions and measurement units. Economic variables from the city-level statistical yearbooks were adjusted for inflation and converted to real terms where necessary. In cases where data inconsistencies arose due to changes in administrative boundaries or definitions, interpolation or imputation techniques were applied.

Survey data from the CMDS were cleaned and weighted to reflect the representativeness of the sample, with missing data points addressed using multiple imputation methods. This ensured that the analysis captured the full diversity of the migrant population.

Spatial analysis was conducted using GIS tools to map the distribution of population, amenities, and economic output across Chinese prefectures. These maps provided a visual representation of the regional disparities that the model seeks to explain. The ArcGIS OD Cost Matrix Tool was instrumental in estimating trade costs, which were then integrated into the model as spatial frictions influencing migration and economic decisions.

The robustness of the model's predictions was tested through sensitivity analysis, where key parameters were varied within plausible ranges to observe their impact on the results. Alternative model specifications were also explored to ensure that the findings were not driven by specific assumptions. The counterfactual analysis, simulating the scenario of *hukou* abolition, was conducted by removing institutional constraints on labor mobility and observing the resulting changes in population distribution, productivity, and economic output. The dynamic-hat algebra method was used to solve the model under these new conditions, allowing for a direct comparison of baseline and counterfactual outcomes.

Appendix E Estimation and Calibration Details

This appendix provides additional details on the estimation procedures, calibration strategy, and construction of quantitative inputs used in the model.

E.1 Elasticities of Amenities

In the model, local amenity levels are endogenously determined by both population density and amenity investment financed through land rents. Specifically, amenities evolve according to:

$$A_t(n) = \bar{A}(n) \cdot \bar{l}_t(n)^{\lambda} \cdot I_t(n)^{\chi}, \tag{E.1}$$

where $\bar{A}(n)$ denotes exogenous geographic amenities, $\bar{l}_t(n)$ is population density, and $I_t(n)$ is amenity investment per capita. Log-linearizing yields:

$$\log A_t(n) = \log \bar{A}(n) + \lambda \log \bar{l}_t(n) + \chi \log I_t(n). \tag{E.2}$$

The specification captures two opposing effects of density on urban livability. Higher density can improve amenities through economies of scale in public-service provision, infrastructure, and cultural activity (Brueckner & Largey, 2008). At the same time, excessive density can generate congestion, environmental degradation, and pressure on local infrastructure (Cavailhes, Gaigné, Tabuchi, & Thisse, 2007).

To empirically estimate λ and χ, I adopt a two-step strategy. First, I decompose amenity levels into exogenous and endogenous components by regressing a composite amenity index on geographic and climatic fundamentals. The fitted values define $\bar{A}(n)$, while the residual captures the portion of amenities responsive to population and investment dynamics.

In the second stage, I estimate the elasticity parameters using a GMM framework. This approach accounts for potential endogeneity in both population density and rent-financed amenity investment. Specifically, I use deep lags of population and predicted rents (based on land supply and location-specific productivity) as instruments. The GMM moments are constructed to match the conditional covariance of residualized amenity outcomes with the instrument set.

E.1.1 Constructing the Amenity Index

The amenity index is constructed using Principal Component Analysis (PCA) applied to standardized indicators from five categories:

1. **Environmental Quality.** Industrial SO_2 emissions (tons/km^2), green coverage rate, arable land (thousand hectares), and noise-affected surface area.
2. **Transportation Infrastructure.** Paved road area per capita, public transit vehicles per 10,000 persons, and taxis per 10,000 persons.
3. **Healthcare.** Hospital beds per capita and physicians per capita.
4. **Education.** Public education expenditure per capita, schools per capita, and full-time teachers per capita.
5. **Cultural Environment.** Number of cinemas per capita and library books per capita.

The first several principal components explaining the majority of cross-city variation are retained to construct the composite amenity index.

E.1.2 Amenity Investment

Amenity investment $I_t(n)$ is proxied using per capita expenditures on city maintenance and environmental protection from the *China City Statistical Yearbooks*. These expenditures are closely linked to local government land-lease revenues and public infrastructure investment.

E.1.3 Exogenous Amenities

To isolate the exogenous component of amenities, I regress the composite amenity index $\log A_t(n)$ on time-invariant geographic characteristics, including:

1. **Topography.** Topographic relief index.
2. **Climate.** Average annual temperature and humidity.
3. **Water Resources.** Share of water bodies in total land area (National Geographic Information Center).

The predicted component defines $\hat{\bar{A}}(n)$, while the residual captures endogenous amenity variation associated with economic activity and local investment.

E.1.4 Estimation of λ and χ

We estimate the relationship between endogenous amenities, population density, and amenity investment using the following specification:

$$\log A_t(n) - \log \hat{\bar{A}}(n) = \lambda \log \bar{l}_t(n) + \chi \log I_t(n) + \epsilon_t(n), \tag{E.3}$$

where $\bar{l}_t(n)$ denotes population density. To address endogeneity concerns (e.g., population density and amenities being jointly determined), we use deep lags of $\bar{l}_t(n)$ and $I_t(n)$ as instruments. Standard errors are clustered at the regional level.

The preferred estimates are:

$$\hat{\lambda} = -0.39, \qquad \hat{\chi} = 0.23.$$

The negative value of λ indicates that congestion effects dominate agglomeration benefits in highly dense urban areas, while the positive value of χ implies that rent-financed public investment significantly improves local amenities.

E.2 Migration Elasticity and Heterogeneous Amenity Preference

Migration decisions depend on expected utility, which includes real income, amenities, migration costs, and idiosyncratic preference shocks. Individual-specific shocks follow a Fréchet distribution governed by the shape parameter γ. Higher values of γ imply more homogeneous preferences and stronger migration responses to spatial utility differences across locations.

Following the spatial equilibrium literature, I interpret γ as the elasticity of migration flows with respect to real wage differentials, conditional on migration costs and destination characteristics.

The migration elasticity is estimated using the following gravity specification:

$$\log M_{n\to j,t} = \gamma \log\left(\frac{w_{j,t}}{w_{n,t}}\right) - \theta \log D_{n,j} + \eta_j n + \eta_j + \eta_t + e_{njt},$$

where:

- $M_{n\to j,t}$ is the observed number of migrants from origin n to destination j at time t,
- $w_{j,t}$ and $w_{n,t}$ are real wages in the destination and origin respectively,
- $D_{n,j}$ is the geodesic distance between n and j,
- η_n, η_j and η_t are destination and time fixed effects.

The estimation uses pooled waves of the China Migrants Dynamic Survey (CMDS). Standard errors are clustered at the origin-destination level.

The estimated migration elasticity is approximately 1.8. Consistent with the spatial equilibrium literature, I set $\gamma = 2$ in the quantitative analysis.

E.3 Agglomeration Effects in Productivity Dynamics

Regional productivity evolves according to:

$$\log(\text{GDPpc}_{it}) = \lambda_1 \log(\text{PopDensity}_{i,t-1}) + \alpha \log(\text{GDPpc}_{i,t-1}) + \epsilon_{it}, \tag{E.4}$$

The parameter λ_1 captures the elasticity of productivity with respect to population density, reflecting localized spillovers through learning, matching, and knowledge diffusion. The parameter α governs productivity persistence.

Several estimation methods are implemented to address potential endogeneity and dynamic panel bias:

- **Fixed Effects (FE)** regression, controlling for unobserved time-invariant regional heterogeneity.
- **Instrumental Variables (IV)** estimation using deep lags (e.g., $L_5.\log(\text{GDPpc})$, $L_5.\log(\text{PopDensity})$) as instruments.

- **Dynamic Panel GMM** estimators with lagged levels as instruments.
- **Nonlinear GMM**

Table E1 summarizes the estimates.

Table E1: **Estimation of λ_1.**

Method	λ_1 (Agglomeration Elasticity)
Fixed Effects (Region and Year)	0.16
IV: L5.lgdppc+ Year FE	0.39
IV: L5.lgdppc and L5.popdens	0.19
IV: L5.lgdppc and L5.popdens + Year FE	0.34
`xtdpdgmm`, lag(2)	0.21
`gmm`, lag(1) instruments	0.021
`gmm`, lag(2) and (3) instruments	0.014

The estimates range from 0.014 to 0.39 depending on the specification and instrument set. Estimates from IV using deep lags with year fixed effects yield relatively high values, suggesting stronger agglomeration externalities. While estimates using GMM with limited lag depth show weaker effects, potentially due to instrument weakness or multicollinearity.

The preferred estimate is $\lambda_1 = 0.21$ based on the difference-GMM specification using second lags as instruments. This estimate is consistent with the broader agglomeration literature and reflects relatively strong urban spillovers during China's period of rapid urbanization (Combes, Duranton, Gobillon, Puga, & Roux, 2012; Ciccone, 2002; Bento, Cropper, Mobarak, & Vinha, 2018).

E.4 Labor Share and Production Parameters

Production combines labor and land according to:

$$Y_t^{\omega}(n) = Z_t^{\omega}(n) L_t^{\omega}(n)^{\iota} H_t^{\omega}(n)^{1-\iota}. \tag{E.5}$$

The labor share parameter ι is estimated using prefecture-level labor compensation and GDP data from the China City Statistical Yearbooks. The resulting estimates imply that labor accounts for the majority of production costs, while land remains quantitatively important due to its central role in urban development and local fiscal finance.

The calibrated value of ι is broadly consistent with estimates in the Chinese macroeconomic literature (typically ranging from 50% to 70%)[8], while also reflecting the importance of land-intensive urban production in rapidly developing metropolitan regions.

E.5 Bilateral Trade Costs

Trade costs are modeled as iceberg costs between prefectures. Following Desmet et al. (2018), bilateral trade costs are constructed using GIS-based travel times across China's on-land transportation network.

[8]Xu, Chen, and Li (2015) find labor shares of 71% at the sectoral level and 58% at the provincial level. Empirical analysis using prefecture-level panel data from 1993 to 2003 suggests an even lower labor share of around 34% after accounting for year and location-fixed effects, which is consistent with what Qi (2015) found using different measurements.

The ArcGIS Origin–Destination Cost Matrix tool is used to compute the fastest travel route between each pair of prefectures using highways, roads, and railways from the year 2000 transportation network. The resulting bilateral travel time between prefectures n and s is denoted $t(n, s)$.

Trade costs are then constructed using:

$$\upsilon(n, s) = 1 + t(n, s)^{0.632}. \tag{E.6}$$

This specification captures increasing marginal trade frictions over distance and is consistent with the empirical trade literature.

For intra-prefectural trade:

$$\upsilon(n, n) = 1.$$

The trade elasticity parameter, which also governs the dispersion of firm-level productivity draws, is calibrated using standard values from the Eaton–Kortum literature rather than estimated directly from the data.

Appendix F Approximation of Equilibrium Allocations Period 0

This section outlines the numerical procedure used to estimate spatial equilibrium allocations at the initial stage of the model (period zero), where population distributions, wages, amenities, and trade flows are jointly determined. Given the nonlinear and interdependent nature of these variables, I implement a fixed-point iterative approach to recover equilibrium values that are consistent with observed data from the year 2000 (period $t = 0$).

The estimation focuses on recovering three central objects in equilibrium: (i) the exogenous amenity component $a(n)$, (ii) the endogenous productivity level, (iii) the population density.

F.1 Initialization and Structure

I initialize the ratio of exogenous amenity component $a(n)$ over the deterministic utility $\tilde{u}_t(n)$ across regions, and set the initial guess for population allocation $l_0(n)$ as a uniform distribution scaled by the total population. The key equations governing the system include: i) endogenous productivity $Z_0(n)$, which depends on wages, land area, and exogenous amenities; ii) a labor demand equation from firms, derived from equilibrium price indices and trade flows. A consistency condition requiring that the population allocation implied by these equations matches the observed population distribution.

F.2 Iterative Algorithm

The solution is obtained via a two-layer fixed-point iteration:

- Inner Loop 1 (Exogenous Amenities $a(n)$): Given initial guesses of $l(n)$, I compute the implied trade-based price indices using the bilateral trade cost matrix and production parameters. These enter into an expression for local expenditure, allowing recovery of the $a(n)$ terms that rationalize trade flows and price levels. Convergence is assessed by the squared deviation between consecutive $a(n)$ vectors.

- Inner Loop 2 (Population Allocation $l(n)$): With $a(n)$ and implied $Z(n)$, I solve for labor allocations that satisfy labor demand, using the functional form derived from the model. The convergence criterion here is the squared deviation between the updated $l(n)$ and the prior guess.

- Outer Loop (Matching to Data): After both inner loops converge, I evaluate the distance between the model-implied population distribution and the observed data. This outer error is defined as the normalized deviation:

$$\text{Error}_{\text{outer}} = \frac{\|H(n) \cdot (l(n) - l_0(n))\|}{\|H(n) \cdot l_0(n)\|}.$$

  The outer loop continues until either the error falls below a specified tolerance threshold or the improvement in the outer error becomes negligible over successive iterations (i.e., convergence stalls).

Appendix G Consumption-Equivalent Welfare Calculation

To evaluate the welfare implications of abolishing the *hukou* system, we compute the present discounted value (PDV) of deterministic utility along the transition path, following an Aiyagari-style aggregation. In each period t, let $\tilde{u}_t(n)$ denote the deterministic component of indirect utility in location n—that is, the component implied by equilibrium prices, wages, amenities, and migration frictions, excluding the i.i.d. taste shocks that rationalize migration probabilities in the discrete-choice migration block. Aggregate deterministic utility in period t is given by:

$$U_t = \sum_{n=1}^{N} \tilde{u}_t(n) \cdot H(n) \cdot \bar{l}_t(n),$$

where $H(n)$ is the land area of location n and $\bar{l}_t(n)$ is the equilibrium labor density.

The PDV of deterministic utility is then:

$$\text{PDV}_U = \sum_{t=1}^{T} \beta^t \, U_t,$$

where $\beta \in (0, 1)$ is the intertemporal discount factor. We compute PDV_U for both the baseline (with *hukou*) and the counterfactual (abolishing *hukou*, i.e., $\phi(h, n) = 1$ for all (h, n)), and measure the welfare change as:

$$\Delta\text{PDV}_U = \frac{\widehat{\text{PDV}}_U - \text{PDV}_U}{\text{PDV}_U}.$$

To express this welfare change in more interpretable terms, we convert it into a consumption-equivalent (CE) welfare measure. Assuming per-period deterministic utility is homothetic in consumption with CRRA curvature ρ and separable from amenities, scaling consumption by a constant factor $(1 + g)$ multiplies period utility by $(1 + g)^{1-\rho}$. This implies:

$$\frac{\widehat{\text{PDV}}_U}{\text{PDV}_U} = (1 + g)^{1-\sigma} \quad \Rightarrow \quad g = \left(\frac{\widehat{\text{PDV}}_U}{\text{PDV}_U} \right)^{\frac{1}{1-\sigma}} - 1, \tag{G.1}$$

where $100 \times g$ is reported as the CE welfare gain in percentage terms.

In our benchmark calibration, we set $\rho = 2$ following standard macroeconomic practice. In this case, the CE welfare gain simplifies to:

$$g = \frac{\text{PDV}^{\text{cf}}}{\text{PDV}^{\text{base}}} - 1. \tag{G.2}$$

Example – Full *Hukou* Abolition Scenario

Table G1: **PDV of Welfare Gains under Full *Hukou* Abolition.**

Horizon	$\textbf{PDV}_{\textbf{base}}$	$\textbf{PDV}_{\textbf{cf}}$	**CE Welfare Gain** g (%)
$t = 2$	14,000,000,000	20,900,000,000	+48.51
$t = 10$	30,800,000,000	47,000,000,000	+52.87

Notes: PDV denotes the present discounted value of aggregate consumption in the baseline (PDV_{base}) and counterfactual full abolition scenario (PDV_{cf}). g represents the percentage change in consumption-equivalent (CE) welfare relative to the baseline equilibrium.

These figures indicate that households would require nearly 49% higher permanent consumption in the baseline to match their welfare in the counterfactual by $t = 2$, and over 52% higher by $t = 10$, highlighting the long-run benefits of removing institutional migration barriers.

This approach isolates the welfare effects of *hukou* abolition that operate through deterministic changes in consumption and amenities, abstracting from transitory or idiosyncratic migration shocks. It ensures that the reported gains reflect permanent, economy-wide improvements in location-specific economic conditions rather than stochastic variation.

Appendix H Sensitivity and Robustness

This subsection examines how the long-run welfare gains from the full abolition of *hukou* barriers (Scenario S0) respond to variation in four key parameters: (i) the amenity–density elasticity (λ), (ii) the productivity–density elasticity (λ_1), (iii) the migration elasticity (γ), and (iv) the amenity–investment elasticity (χ).

The amenity–density elasticity λ captures congestion effects from high urban density; Chinese megacities such as Beijing and Shanghai face some of the world's highest population densities, making this channel potentially strong. The productivity–density elasticity λ_1 reflects agglomeration spillovers, shaped by China's industrial clustering in manufacturing and services, especially in coastal growth poles. The migration elasticity γ measures the responsiveness of labor flows to utility differences across locations. In the model, migration costs arise from two sources—institutional frictions from the *hukou* system and distance-based costs—so γ governs how strongly workers reallocate once those frictions change. Finally, the amenity–investment elasticity χ reflects how effectively local governments convert land-lease revenues into public services and infrastructure—an important channel in China, where land finance accounts for a substantial share of local fiscal revenue.

For each parameter, we simulate a "Low" and "High" value corresponding to a one-standard-error decrease or increase from the baseline estimate (or an equivalent calibrated range where standard errors are unavailable), holding all other parameters fixed. The ranges are summarized in Table H1.

While a full robustness check across all counterfactual scenarios is left for future work, the S0 results provide an informative benchmark. Table H2 shows that the welfare gains from

Table H1: **Parameter Ranges for Sensitivity Tests.**

Parameter	Baseline	Low	High
Amenity–density (λ)	-0.39	-0.50	-0.20
Productivity–density (λ_1)	0.21	0.15	0.30
Migration elasticity (γ)	2.00	1.00	3.00
Amenity–investment (χ)	0.23	0.15	0.35

Table H2: **Sensitivity of Long-Run CE Welfare Gains at $t = 10$.**

Scenario	Baseline	Low λ	High λ	Low λ_1	High λ_1
S0: Full abolition	52.87	48.67	56.40	49.80	58.80
Scenario	Low γ	High γ	Low χ	High χ	
S0: Full abolition	45.30	61.00	48.20	59.99	

Notes: "Low" and "High" denote a one-standard-error decrease or increase from the baseline parameter value (or an equivalent calibrated range where standard errors are unavailable). CE values are percentage changes relative to the baseline equilibrium. $t = 10$ corresponds to the long-run equilibrium after reform.

complete *hukou* abolition are most sensitive to γ and χ, highlighting the importance of migration responsiveness and the role of public-investment-driven amenity improvements. By contrast, varying λ or λ_1 within empirically reasonable bounds changes the magnitudes more moderately, suggesting that congestion and agglomeration spillovers, while important, are less decisive in determining aggregate gains in this policy experiment.

Overall, these results suggest that the main policy message—nationwide *hukou* abolition yields large long-run welfare gains—remains qualitatively robust to reasonable parameter variation.

Appendix I Prefectures in China

China's administrative structure is hierarchical, with the country divided into several levels of government, each with its own jurisdictions and responsibilities. One of the key administrative divisions is the prefecture, an important unit that lies between the provincial and county levels. Prefectures serve as crucial links in the governance chain, managing large areas that typically encompass multiple counties, districts, and even cities. These prefectures play a vital role in implementing policies, managing resources, and serving as hubs for economic and social development.

Prefecture-level cities are particularly significant in China's urban hierarchy. They usually include a central urban area and its surrounding rural regions, with the central city acting as the administrative and economic heart. The number of prefecture-level cities in China has evolved over time due to administrative changes such as mergers, abolitions, and the establishment of new cities. As of 2000, there were 333 prefecture-level units in China, but after various adjustments, this number was streamlined to 313, which are the focus of the study in this paper. These prefectures are diverse, encompassing economically advanced coastal cities, less-developed inland regions, and a wide range of geographic, cultural, and economic environments.

Overlaying this administrative landscape is China's *hukou* system, a household registration policy that has been in place since the 1950s. The *hukou* system categorizes Chinese citizens based on their place of residence and birth, effectively tying individuals to a specific location and controlling their access to various social services such as education, healthcare, and housing. Initially designed to manage rural-to-urban migration and ensure social stability, the *hukou* system has become a major determinant of social mobility in China.

Appendix J Institutional Background: The *Hukou* System

This appendix provides additional institutional context on the *hukou* system in China. It traces the system's origins, major policy adjustments, and the decentralized governance structure that continues to shape migration outcomes today.

J.1 Historical Origins: 1950s–1970s

The *hukou* system was formally established in 1958 under the Regulations on Household Registration of the People's Republic of China. Its original purpose was to control rural-to-urban migration during the early stages of industrialization, when the state sought to ensure stable grain supply, prevent overcrowding in urban areas, and protect limited public services. By tying access to education, healthcare, housing, and food rations to an individual's registered *hukou* location, the system effectively bound people to their birthplace. For more than two decades, internal migration required explicit state authorization or employer sponsorship, and unauthorized relocation often led to fines or forced return. *Hukou* thus became a cornerstone of China's planned economy, deeply shaping patterns of labor allocation and urbanization.

J.2 Reforms and Decentralization: 1970s–1990s

The launch of the Reform and Opening policy in 1979 marked the first step toward loosening *hukou* restrictions. Authority over *hukou* transfers was gradually delegated from the central Ministry of Public Security to provincial and local governments, creating scope for regional variation. Temporary residence permits (zanzhuzheng) were introduced in the early 1980s, allowing rural migrants to relocate for work without gaining full local registration. Meanwhile, rapid industrial expansion in coastal provinces created unprecedented demand for labor, fueling large flows of rural migrants to cities. Although these changes facilitated mobility, access to public services remained largely restricted to registered residents, reinforcing the divide between locals and migrants.

J.3 Gradual Adjustment: 2000s

Policy adjustments in the 2000s reflected growing recognition of migrants' role in urban economies. In 2001, the State Council encouraged mid-sized cities to relax *hukou* criteria in order to absorb surplus rural labor. A landmark change came in 2003 with the abolition of the custody and repatriation system, which had allowed local authorities to detain and forcibly return migrants without *hukou*. In the mid-2000s, several provinces experimented with pilot programs that extended partial access to schooling for migrant children or included migrants in limited social insurance schemes. Nonetheless, coverage remained uneven and contingent on local fiscal capacity. By 2010, the floating population had exceeded 200 million, underscoring both the scale of migration and the persistence of institutional exclusion.

J.4 Recent Reforms and Persistent Fragmentation: 2010–present

The 2010s brought renewed calls for *hukou* reform, culminating in the National New-Type Urbanization Plan of 2014. This plan encouraged cities with fewer than three million residents to remove *hukou* restrictions entirely, while instructing larger cities to implement points-based or quota systems. In 2016, the rural–urban *hukou* distinction was formally abolished, but locality-based restrictions remained in place, meaning that migrants without local registration continued to face barriers to public services. Today, Tier-1 metropolises such as Beijing, Shanghai, Shenzhen, and Guangzhou maintain highly restrictive systems, often tying eligibility to educational attainment, housing ownership, or tax contributions. By contrast, many smaller cities have significantly relaxed criteria to attract labor and investment.

J.5 Fragmented Governance

The decentralization of *hukou* management has produced a fragmented institutional landscape. Local governments have substantial discretion to set admission rules in line with their fiscal capacity, demographic pressures, and development strategies. This discretion has resulted in starkly different regimes across cities. Shanghai employs a transparent points-based system, rewarding education, tax contributions, and employment in skilled sectors, while Beijing operates a more opaque employer-linked quota system with limited transparency. Smaller cities, facing depopulation or labor shortages, often provide relatively straightforward pathways to *hukou* acquisition. The outcome is a patchwork of local rules that determine migrants' settlement opportunities in highly uneven ways.

This fragmented governance structure directly motivates the counterfactual policy scenarios examined in the main text. The three modeled interventions—nationwide abolition (S0), selective reforms in high-barrier cities (S1), and reforms concentrated in major economic hubs (S2)—mirror the diversity of current institutional practices and allow for an evaluation of their long-run implications for labor mobility, productivity, and welfare.